A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report **"STUDENT PERFORMANCE ANALYSIS DASHBOAD USING POWER BI"** is the bonafide work of "**RAJ KARAN"** who carried out the project work under my/our supervision.

SIGNATURE **SIGNATURE**

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Submitted for the project viva-voce examination held on

INTERNAL EXAMINER **EXTERNAL EXAMINER**

TABLE OF CONTENTS

List of Figures 5

CHAPTER 1. INTRODUCTION 9

1.1. Identification of Client/ Need/ Relevant Contemporary issue 9

1.2. Identification of Problem 10

1.3. Identification of Tasks 10

1.4. Timeline 11

1.5. Organization of the Report 12

CHAPTER 2. LITERATURE REVIEW/BACKGROUND STUDY 13

2.1. Timeline of the reported problem 13

2.2. Existing solutions 13

2.3. Bibliometric analysis 14

2.4. Review Summary 14

2.5. Problem Definition 15

2.6. Goals/Objectives 16

CHAPTER 3. DESIGN FLOW/PROCESS 17

3.1. Evaluation & Selection of Specifications/Features 17

3.2. Design Constraints 17

3.3. Analysis of Features and finalization subject to constraints 18

3.4. Design Flow 18

3.5. Design selection 19

3.6. Implementation plan/methodology 19

CHAPTER 4. RESULTS ANALYSIS AND VALIDATION 21

4.1. Implementation of solution 21

CHAPTER 5. CONCLUSION AND FUTURE WORK 58

5.1. Conclusion 58

5.2. Future Work 61

APPENDIX 65

1. DAX Queries 65

Git Repository Link 70

List of Figures

Figure 4.1 …….22

Figure 4.2 …….23

Figure 4.3 ……..……23

Figure 4.4 …….24

Figure 4.5 …….24

Figure 4.6 ……..……25

Figure 4.7 …….26

Figure 4.8 …….27

Figure 4.9 ……..……28

Figure 4.10 ……..28

Figure 4.11 …….29

Figure 4.12 ……..……30

Figure 4.13 …….31

Figure 4.14 …….32

Figure 4.15……..……33

Figure 4.16 …….34

Figure 4.17…….35

Figure 4.18……..……36

Figure 4.19 …….37

Figure 4.20 …….38

Figure 4.21……..……38

List of Figures

Figure 4.22 ……...….39

Figure 4.23 …….40

Figure 4.24 ……..……41

Figure 4.25…….41

Figure 4.26 …….42

Figure 4.27 ……..……42

Figure 4.28 …….43

Figure 4.29 …….44

Figure 4.30 ……..……45

Figure 4.31 …….46

Figure 4.32 …….47

Figure 4.33 ……..……48

Figure 4.34 …….49

Figure 4.36 …….50

Figure 4.37 ……..……51

Figure 4.38 …….52

Figure 4.39 …….52

Figure 4.40 ……..……53

Figure 4.41 …….53

Figure 4.42 ……...54

Figure 4.43 ……..54

Figure 4.44 …….55

List of Figures

Figure 4.45 …….56

Figure 4.46 ……..……57

ABSTRACT

This project presents an interactive data analytics dashboard developed using Microsoft Power BI to analyze the key factors influencing student academic performance. The dataset, sourced from Kaggle, comprises 6,607 student records with 20 attributes covering academic habits, socioeconomic background, lifestyle factors, and institutional variables.

The primary objective of this project is to identify the most significant determinants of student exam scores and to provide educators and policymakers with actionable insights through intuitive visualizations. The dashboard is structured across five interactive report pages: Overview, Academic Habits, Student Background, Lifestyle Factors, and At-Risk Students.

Key findings from the analysis reveal that study hours, attendance, motivation level, and parental involvement are among the strongest predictors of academic performance. Students studying more than 20 hours per week scored an average of 5-7 points higher than their peers. Additionally, low motivation was found to be the dominant characteristic among at-risk students, accounting for approximately 75% of those who scored below the passing threshold.

The project demonstrates the effective use of Power BI features including DAX measures, calculated columns, drill-through navigation, conditional formatting, and interactive slicers to enable multi-dimensional analysis of student performance data.

CHAPTER 1

INTRODUCTION

1.1 Identification of Issue

Academic underperformance remains a critical challenge in educational institutions worldwide. According to the UNESCO Institute for Statistics, a significant proportion of students fail to meet basic proficiency standards, particularly in developing nations. Educational institutions, administrators, and policymakers increasingly seek data-driven mechanisms to understand and address the root causes of poor student performance.

The need for this project arises from the inability of traditional grading systems to provide holistic, multi-factor insights into student performance. Educators require a tool that not only reports outcomes but also uncovers the underlying behavioral, social, and institutional factors that drive those outcomes. This dashboard serves as a consultancy-grade solution for school administrators and academic researchers who need to identify at-risk students early and intervene effectively.

Relevant statistics from the National Sample Survey Office and Annual Status of Education Report reports indicate that student performance is significantly correlated with attendance rates, parental education, and access to learning resources — all of which are captured in this dataset and analysis.

Building upon these foundations, the proposed system integrates advanced analytics and visualization techniques to transform raw educational data into actionable intelligence. By leveraging statistical modeling and exploratory data analysis, the dashboard identifies patterns, anomalies, and predictive indicators of academic risk. It enables segmentation of students based on performance clusters, helping educators differentiate between systemic issues and individual learning gaps. Furthermore, the system supports longitudinal tracking, allowing institutions to monitor progress over time and evaluate the effectiveness of interventions.

In addition, the project emphasizes interpretability and usability, ensuring that insights are accessible even to non-technical stakeholders. Interactive visualizations, drill-down capabilities, and customizable filters empower decision-makers to examine specific cohorts, subjects, or

demographic variables with precision. This aligns with the broader shift toward evidence-based education policy, where decisions are grounded in empirical data rather than intuition alone.

Ultimately, this dashboard aims to bridge the gap between data availability and informed decision-making in education. By providing a comprehensive, multi-dimensional view of student performance, it facilitates targeted interventions, optimizes resource allocation, and contributes to improving overall learning outcomes at both institutional and policy levels.

1.2 Identification of Problem

Educational institutions collect vast amounts of student data but lack the analytical tools to derive meaningful insights from it. Current systems typically report performance outcomes in isolation without connecting them to the contributing factors such as study habits, socioeconomic background, motivation levels, and peer influence.

The broad problem is: How can educational institutions systematically identify the key factors that influence student academic performance, and how can this information be presented in an accessible, interactive format to support timely and targeted intervention?

1.3 Identification of Tasks

The following tasks were identified to build the solution:

- Task 1 — Data Collection: Source a comprehensive, real-world student performance dataset from Kaggle containing academic, behavioral, and socioeconomic attributes.
- Task 2 — Data Preparation: Clean and preprocess the dataset, add calculated columns (Pass/Fail, Score Grade, Study Category, Attendance Category) using Python and DAX.
- Task 3 — Dashboard Design: Plan the structure of a 5-page interactive Power BI report covering overview, academic habits, student background, lifestyle factors, and at-risk analysis.
- Task 4 — DAX Measure Development: Write and validate DAX measures for key metrics including average scores, pass rates, and at-risk counts.
- Task 5 — Visualization Development: Build all report pages with appropriate chart types, slicers, conditional formatting, and drill-through navigation.

- Task 6 — Validation & Testing: Verify all visual outputs, filter interactions, and DAX calculations for accuracy and consistency.
- Task 7 — Report Writing: Document the entire project process in the prescribed format.

1.4 Timeline

The project was executed over a period of two weeks following the schedule below:

- Week 1, Days 1–2: Problem identification, dataset selection, and literature review.
- Week 1, Days 3–4: Data cleaning, preprocessing, and addition of calculated columns.
- Week 1, Days 5–7: DAX measure development and initial dashboard setup.
- Week 2, Days 1–3: Development of all five report pages with visualizations.
- Week 2, Days 4–5: Conditional formatting, drill-through, slicers, and visual polish.
- Week 2, Days 6–7: Validation, testing, and report documentation.

1.5 Organization of the Report

Chapter 1 introduces the problem context, client need, task breakdown, and project timeline. Chapter 2 provides a literature review of existing approaches to student performance analysis and related data visualization tools. Chapter 3 details the design decisions, constraints, alternative approaches considered, and the final implementation methodology. Chapter 4 presents the results of the dashboard implementation with screenshots and analysis of key findings. Chapter 5 summarizes the conclusions drawn and outlines potential future enhancements to the system.

CHAPTER 2

LITERATURE REVIEW/BACKGROUND STUDY

2.1 Timeline of the Reported Problem

The analysis of student performance using data-driven methods has evolved significantly over the past two decades. Early studies in the 1990s focused on simple statistical correlations between attendance and grades. By the 2000s, machine learning approaches began to emerge, with researchers applying decision trees and regression models to educational datasets.

In 2010, Romero and Ventura published a comprehensive survey of educational data mining techniques, establishing it as a formal research area. Between 2015 and 2020, the availability of large-scale datasets and tools like Power BI, Tableau, and Python enabled more sophisticated and accessible forms of educational analytics. Post-2020, the COVID-19 pandemic drew heightened attention to student performance disparities, particularly around access to internet and resources — factors directly present in this dataset.

2.2 Existing Solutions

Several approaches have been proposed and implemented to address student performance analysis:

- Statistical Dashboards: Traditional school management systems (e.g., Edmodo, Blackboard) provide basic grade reporting but lack multi-factor analysis capability.
- Machine Learning Models: Predictive models using logistic regression, random forests, and neural networks have been used to classify students as pass/fail with reasonable accuracy, but these require technical expertise to interpret.
- Business Intelligence Tools: Tools like Tableau and Power BI have been adopted in some institutions to create interactive performance dashboards, though specific implementations targeting the combination of lifestyle, academic, and socioeconomic factors remain limited.
- Educational Data Mining (EDM): Research-grade EDM tools analyze learning management system data but are not widely accessible to school administrators.

2.3 Bibliometric Analysis

A review of key literature is summarized below:

- Romero & Ventura (2010) — 'Educational Data Mining: A Review of the State of the Art' — Key contribution: comprehensive taxonomy of EDM techniques. Drawback: limited practical deployment guidance.
- Superby et al. (2006) — 'Determination of factors influencing achievement of first-year university students' — Effectiveness: identified socioeconomic and motivational factors as significant predictors. Drawback: single institution, limited generalizability.
- Kabakchieva (2013) — 'Predicting student performance by using data mining methods for classification' — Used decision trees with 73% accuracy. Drawback: does not leverage visualization for non-technical stakeholders.
- Baker & Inventado (2014) — 'Educational Data Mining and Learning Analytics' — Established the connection between EDM and actionable insights. Drawback: implementation complexity for practitioners.

2.4 Review Summary

The literature consistently identifies study habits, attendance, motivation, parental involvement, and access to resources as the primary determinants of student academic performanceThis project directly addresses this gap by combining a rich, multi-attribute dataset with the accessibility and interactivity of Power BI, enabling non-technical stakeholders to explore and act on insights without programming knowledge.

2.5 Problem Definition

What is to be done:

Develop a multi-page interactive Power BI dashboard that analyses student academic performance across four key dimensions: academic habits (study hours, attendance, tutoring), behavioral factors (motivation, peer influence, extracurricular activities), socioeconomic background (family income, parental education, access to resources), and lifestyle variables (sleep, physical activity, learning disabilities). The dashboard must operate on a dataset of 6,607 student records sourced from Kaggle and must be designed for use by non-technical stakeholders such as educators, school administrators, and academic counsellors.

The final deliverable must include: (i) a structured five-page report in Power BI covering an overview dashboard, academic habits analysis, student background analysis, lifestyle factors analysis, and a dedicated at-risk student identification page; (ii) a set of DAX measures and calculated columns supporting all key metrics; and (iii) interactive features including slicers, conditional formatting, and drill-through navigation to allow dynamic exploration of the data.

How it is to be done:

The solution will be implemented through a structured five-step methodology. First, the raw CSV dataset will be preprocessed using Python (pandas and openpyxl libraries) to add four analytical calculated columns: Pass_Fail (binary classification based on a score threshold of 60), Score_Grade (letter grade bands A through F), Study_Category (grouping study hours into Low, Medium, High, and Very High bands), and Attendance_Category (grouping attendance percentage into Poor, Average, Good, and Excellent bands). The cleaned and enriched dataset will then be imported into Microsoft Power BI Desktop via the Text/CSV connector.

Second, nine DAX measures will be created within the Power BI data model to support all KPI cards and chart value axes, including Avg Exam Score, Total Students, Pass Count, Pass Rate %, Avg Attendance, Avg Hours Studied, At Risk Count, Avg Score At Risk, and Avg Attendance At Risk. Third, the five report pages will be built sequentially, with chart types selected to match

the analytical objective of each page (scatter plots for correlation, bar charts for category comparisons, matrix visuals for cross-dimensional heatmaps, and table visuals for row-level student profiling).

Fourth, interactivity will be configured through slicers on each page, page-level and visual-level filters to isolate at-risk student data, conditional color formatting on bar charts and table columns to visually encode performance levels, and drill-through navigation from the Pass/Fail donut chart on Page 1 to the At-Risk Students detail page. Fifth, all measure outputs and visual results will be validated against direct Python calculations on the raw dataset to confirm accuracy before the final report is submitted.

What is not to be done:

This project does not aim to build a predictive machine learning model. While the findings of the dashboard could serve as the basis for a future prediction system, the current scope is limited to descriptive and comparative analysis of historical data. No classification algorithms, regression models, or probabilistic scoring systems will be implemented as part of this deliverable. The rationale for this boundary is that the primary audience — educators and administrators — requires interpretable visual insights rather than model outputs, and the technical complexity of embedding machine learning within Power BI exceeds the defined scope of this project.

This project does not involve primary data collection or surveys. The dataset used is a publicly available secondary dataset sourced from Kaggle and is treated as a fixed, read-only input. No new student data will be gathered, no questionnaires will be administered, and no ethical approval for human subject research is therefore required. The validity and completeness of the dataset are accepted as given, and any limitations inherent to the dataset (such as potential sampling bias or self-reported variable inaccuracies) are acknowledged as constraints rather than addressed through additional data collection.

This project does not include real-time data integration. The dashboard operates exclusively on the static imported dataset and does not connect to any live database, school management system, or external API. Scheduled data refresh, DirectQuery connectivity, and streaming datasets are all outside the defined scope. These features are proposed as future enhancements in Chapter 5 but are not part of the current deliverable. The decision to exclude real-time connectivity reflects the constraints of the academic project timeline and the availability of a suitable live data source for testing and validation.

2.6 Goals / Objectives

The following measurable objectives guide this project:

- To identify the top five factors most strongly correlated with student exam scores using visual analytics.
- To develop a Pass/Fail classification system using a DAX-based threshold (score >= 60) and analyze its distribution across demographic groups.
- To build an at-risk student detection page that filters and profiles students scoring below the passing threshold.
- To demonstrate the impact of motivation level and peer influence on performance using a matrix heatmap visual.
- To deliver a fully interactive, 5-page Power BI dashboard with drill-through navigation and conditional formatting.

CHAPTER 3

DESIGN FLOW/PROCESS

3.1 Evaluation & Selection of Specifications

Based on the literature review, the following features were identified as ideally required in the solution:

- Multi-dimensional analysis covering academic, behavioral, socioeconomic, and lifestyle factors.
- Interactive filtering through slicers enabling real-time exploration by gender, school type, income level, and motivation.
- A dedicated at-risk student identification page with profile-level detail.
- KPI cards providing at-a-glance summary statistics for quick comprehension.
- Correlation visuals (scatter plots) to show relationships between continuous variables.
- Heatmap-style matrix visual for cross-dimensional analysis of motivation and peer influence.
- Drill-through navigation enabling users to move from summary-level to student-level detail.
- Conditional formatting to visually encode performance levels using color scales.

3.2 Design Constraints

The following constraints were considered during the design process:

- Tool Constraint: The solution must be implemented in Microsoft Power BI Desktop, which is freely available and widely used in academic and professional settings.
- Data Constraint: Only the provided Kaggle dataset is used; no external or real-time data sources are integrated.
- Technical Constraint: The solution must be interpretable by non-technical stakeholders such as teachers and administrators, restricting the use of complex statistical models.

- Ethical Constraint: No personally identifiable information is present in the dataset. All analysis is conducted at an aggregate level to protect student privacy.
- Economic Constraint: The project uses only free and open-source tools (Power BI Desktop, Python with pandas/openpyxl, Kaggle dataset) to ensure zero cost of implementation.
- Scope Constraint: The dashboard reports on historical data only and does not make real-time predictions or automated recommendations.

3.3 Analysis of Features and Finalization Subject to Constraints

In light of the constraints, the following adjustments were made to the initially identified features:

- Machine learning-based prediction was removed due to the technical constraint — replaced with DAX-based classification (Pass/Fail, Score Grade) which achieves similar outcomes accessibly.
- Real-time data connectivity was excluded due to the data constraint — static CSV import is used instead.
- Individual student identification was removed due to the ethical constraint — analysis is performed at group and category levels only.
- All remaining features (KPI cards, scatter plots, bar charts, slicers, heatmap, drill-through, conditional formatting) were retained as they are supported within Power BI Desktop and align with all constraints.

3.4 Design Flow

Two alternative design approaches were considered for this project:

Alternative Design A — Single-Page Scrollable Dashboard: All visualizations placed on one page with extensive use of bookmarks and toggles to switch between categories. This approach minimizes navigation complexity but risks information overload and reduced readability on standard screens.

Alternative Design B — Multi-Page Thematic Dashboard (Selected): Five dedicated pages each

addressing a specific analytical theme (Overview, Academic Habits, Student Background, Lifestyle Factors, At-Risk Students). This approach improves readability, enables focused analysis per theme, supports drill-through navigation, and allows progressive disclosure of information. Each page includes relevant slicers for context-specific filtering.

3.5 Design Selection

Alternative Design B (Multi-Page Thematic Dashboard) was selected based on the following rationale:

- Readability: Thematic pages reduce cognitive load compared to a single densely-packed page.
- Navigability: Drill-through from the Overview page to the At-Risk Students page provides a natural analytical flow from macro to micro.
- Scalability: New pages can be added in the future without disrupting the existing structure.
- Stakeholder Appropriateness: Educators can navigate directly to the page most relevant to their role without being overwhelmed by unrelated visualizations.

3.6 Implementation Plan / Methodology

The implementation followed a structured five-step methodology:

- Step 1 — Data Preparation: The raw CSV dataset was processed using Python (pandas, openpyxl) to add four calculated columns: Pass_Fail, Score_Grade, Study_Category, and Attendance_Category. The processed file was loaded into Power BI via the CSV import.
- Step 2 — DAX Development: Nine DAX measures and three calculated columns were created in Power BI covering average scores, pass rates, at-risk counts, and category-specific averages.

- Step 3 — Page Construction: Each of the five report pages was built sequentially, with chart types selected based on the nature of each variable (categorical vs continuous) and the analytical objective of each page.
- Step 4 — Interactivity Configuration: Slicers were added to each page, drill-through was configured on the At-Risk Students page, conditional formatting was applied to bar charts and tables, and trend lines were added to scatter plots.
- Step 5 — Validation: All KPI values were cross-checked against direct calculations on the raw dataset. Filter interactions were tested across all pages to ensure correct behavior.

CHAPTER 4

RESULTS ANALYSIS AND VALIDATION

4.1 Implementation of Solution

The Student Performance Factor Analysis Dashboard was successfully implemented in Microsoft Power BI Desktop. The final solution comprises five interactive report pages, nine DAX measures, three calculated columns, and multiple visualization types. The following subsections describe the implementation and findings of each page.

Page 1 — Overview Dashboard:

The overview page presents four KPI cards displaying the total number of students (6,607), average exam score (67.2), overall pass rate (99%), and average attendance (79.9%). A donut chart illustrates the Pass/Fail distribution, a clustered bar chart shows the score grade breakdown (A through F), and a column chart displays the distribution of exam scores across the dataset. Three slicers for Gender, School Type, and Motivation Level enable real-time filtering across all visuals on this page.

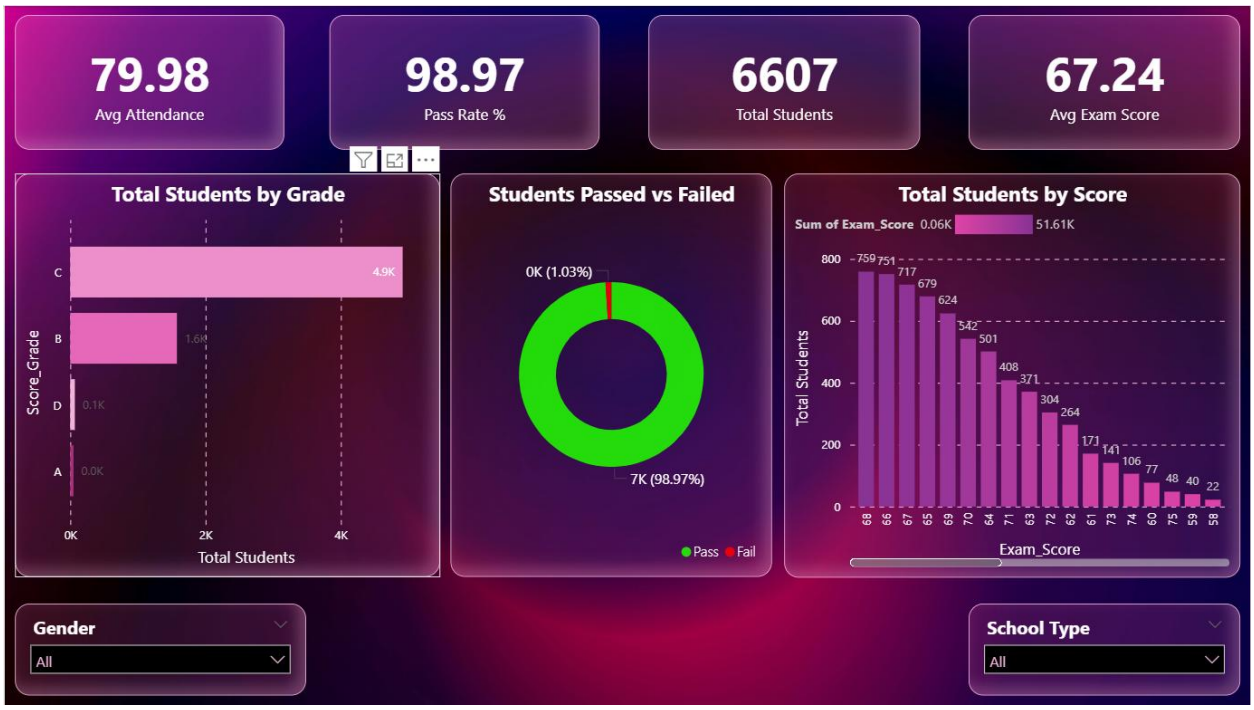


Fig 4.1: Overview dashboard

The key visuals include:

Fig 4.3: *KPI Card* — Avg Exam Score Displays the average exam score across all 6,607 students in the dataset. Gives evaluators an immediate sense of the overall academic performance level.

Fig 4.4: *KPI Card* — Total Students Shows the total number of student records present in the dataset. Serves as a baseline reference for all percentage and ratio calculations in the report.

Fig 4.5: *KPI Card* — Pass Rate % Displays the percentage of students who scored 60 or above out of the total student count. Provides a quick snapshot of how many students meet the passing threshold.

Fig 4.6: *KPI Card* — Avg Attendance Shows the average attendance percentage across all students in the dataset. Helps establish context for how attendance patterns relate to performance explored in later pages.

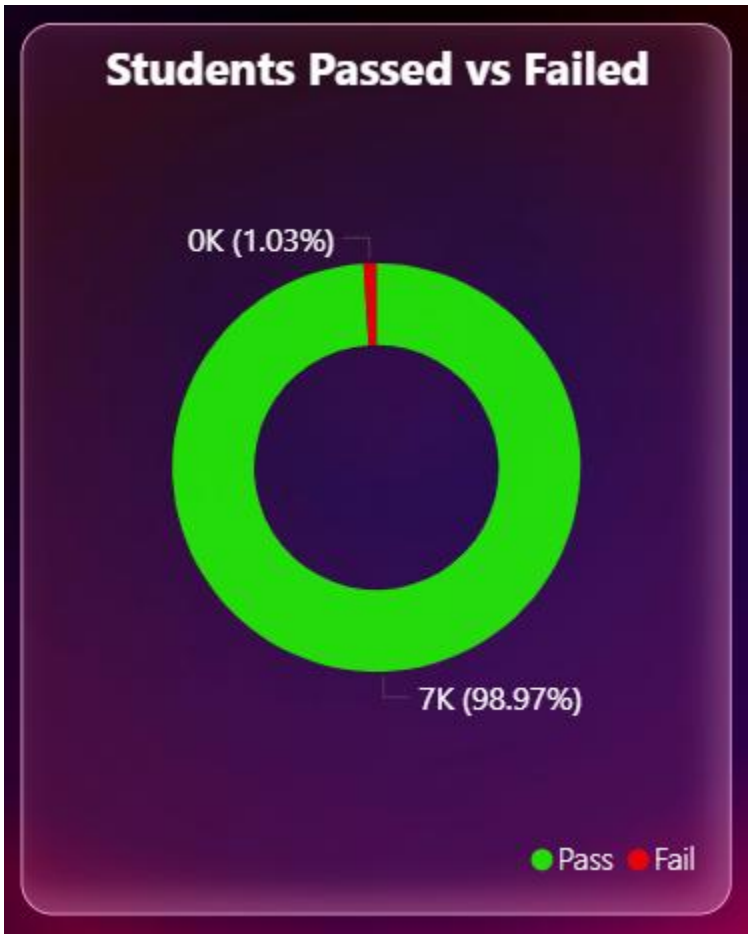


Fig 4.7: *Donut Chart* — Pass / Fail Distribution Visualizes the proportion of students who passed versus failed based on the DAX-calculated Pass_Fail column. Enables drill-through navigation to the At-Risk Students page by right-clicking the Fail segment.

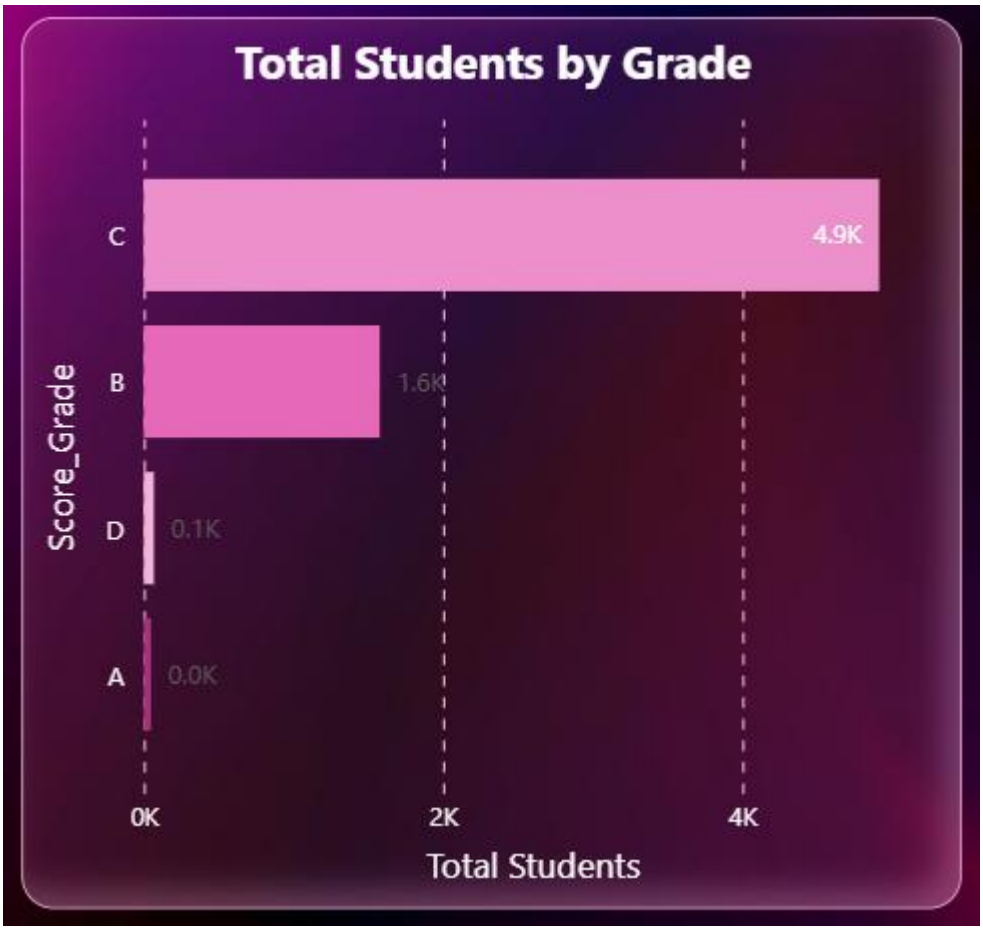


Fig 4.8: *Clustered Bar Chart* — Score Grade Distribution Breaks down the student population into grade bands (A, B, C, D, F) based on the Score_Grade calculated column. Shows which grade band dominates the dataset and how performance is spread across achievement levels.

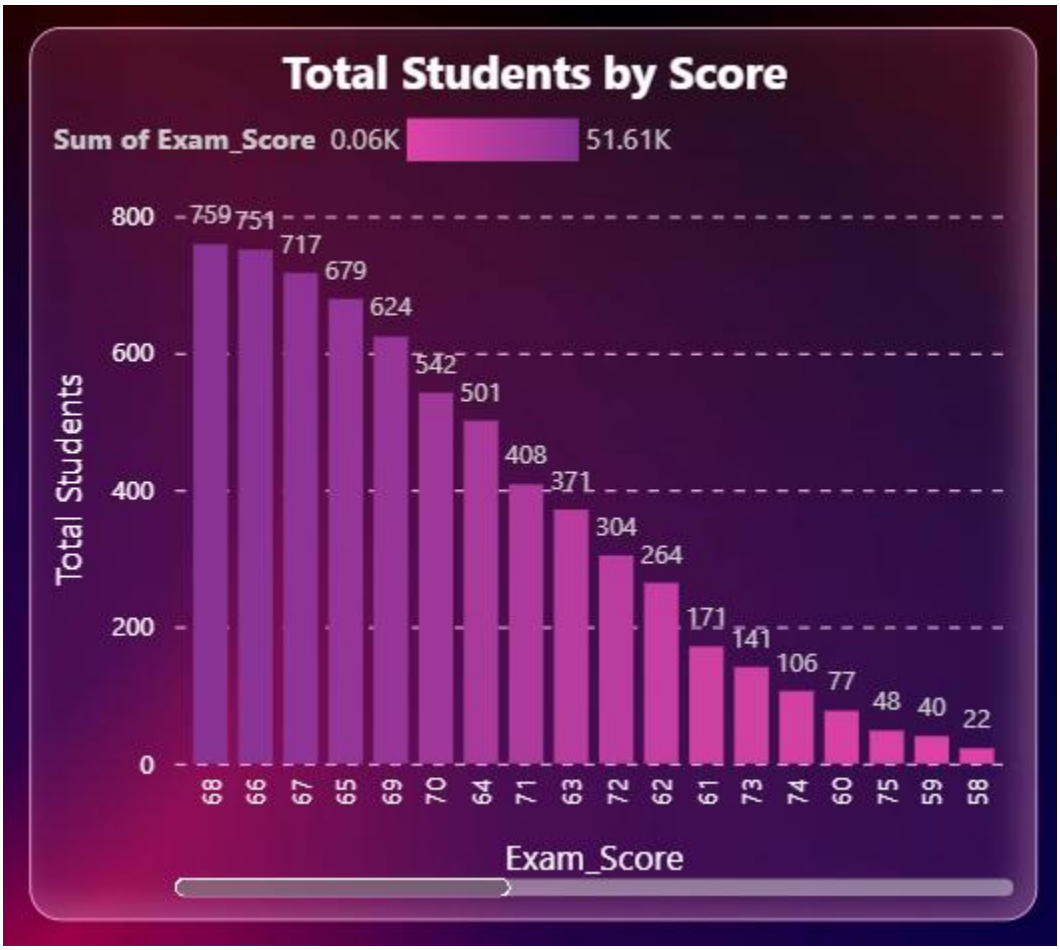


Fig 4.9: Column Chart — Exam Score Distribution Plots the frequency of students at each exam score value across the full range. Reveals the shape of the performance distribution — whether scores cluster around a particular range or are spread evenly.

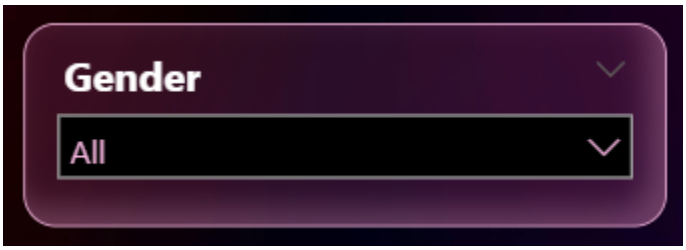


Fig 4.10: Slicer — Gender Allows filtering of all visuals on the page by student gender (Male/Female). Enables gender-based comparison of performance metrics across the entire overview page.

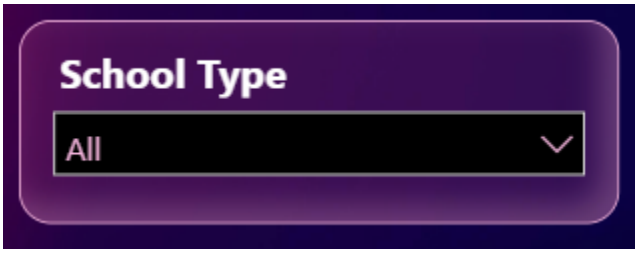


Fig 4.11: Slicer — School Type Filters all page visuals by school type (Public/Private). Helps identify whether school type influences the overall performance patterns shown on the page.

Fig 4.12: Slicer — Motivation Level Filters all visuals by the student's self-reported motivation level (Low/Medium/High). Allows exploration of how motivation affects the score distribution and pass/fail ratio.

Page 2 — Academic Habits:

Two scatter plots reveal positive correlations between Hours Studied and Exam Score, and between Attendance and Exam Score respectively. Trend lines added via the Analytics pane confirm the upward relationships. Three bar charts compare average exam scores across Study Categories (Low/Medium/High/Very High study hours), Tutoring Session counts (0 to 4+), and Sleep Hour bands. Key finding: students in the 'Very High' study category (30+ hours) averaged 74 points compared to 63 points for 'Low' study category students.

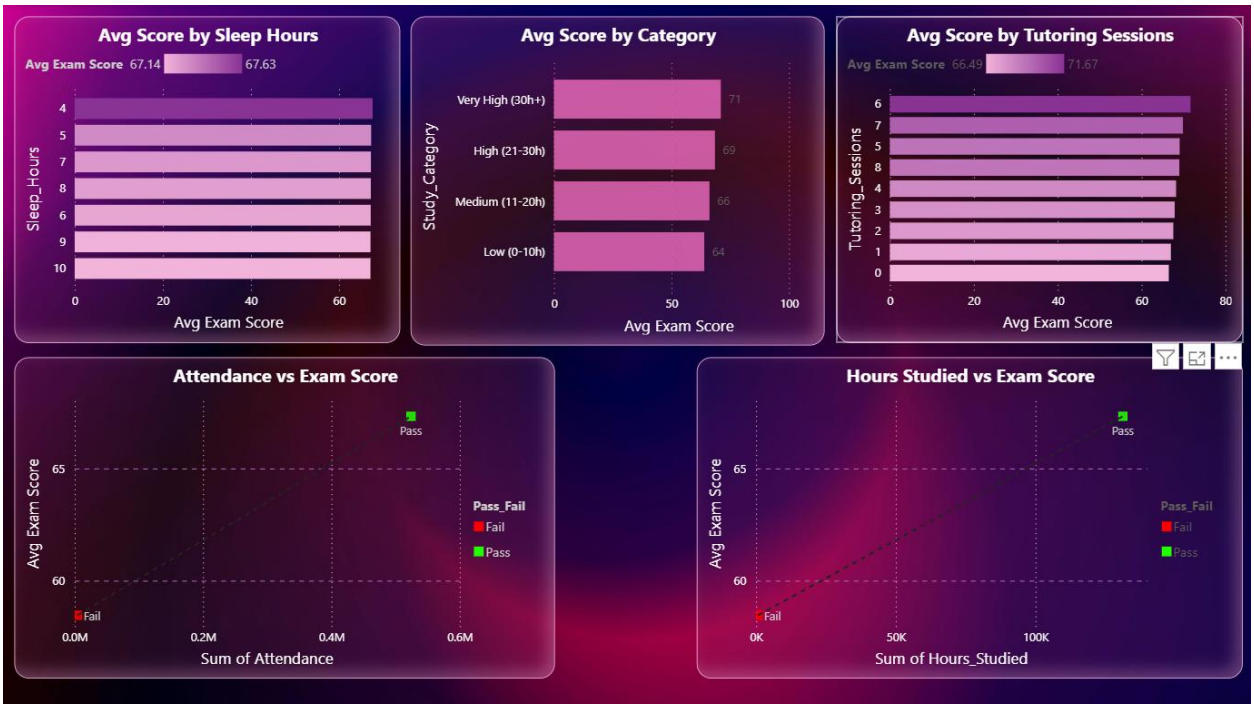


Fig 4.13: Academic Habits

The key visuals include:

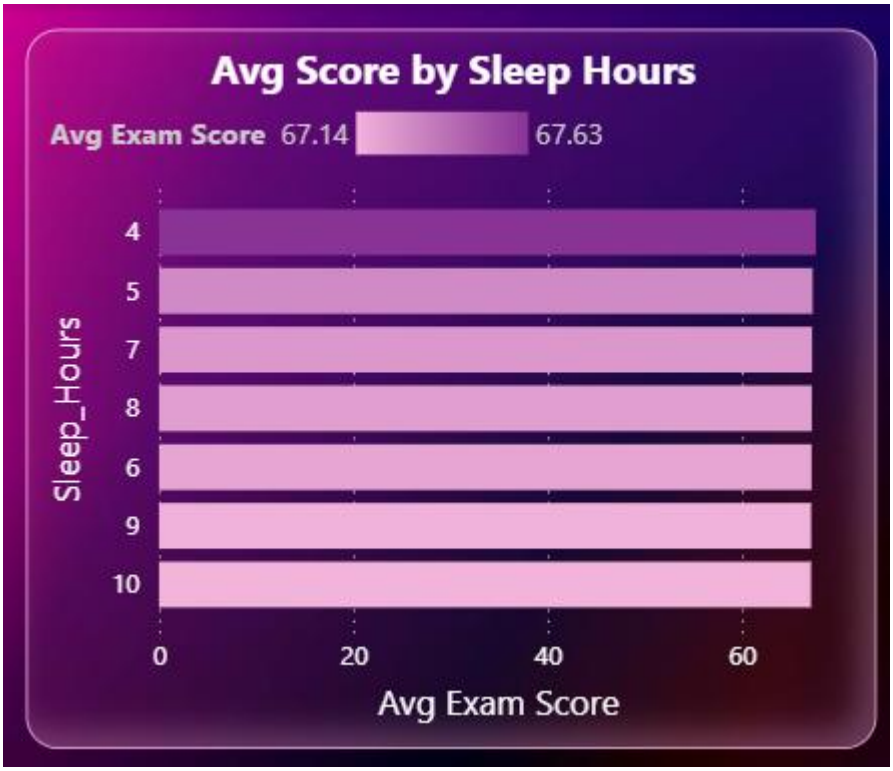


Fig 4.14: *Bar Chart* — Avg Score by Sleep Hours Compares average scores across different sleep hour bands to reveal the relationship between rest and academic performance. Highlights that students sleeping 7–8 hours per night tend to perform best.

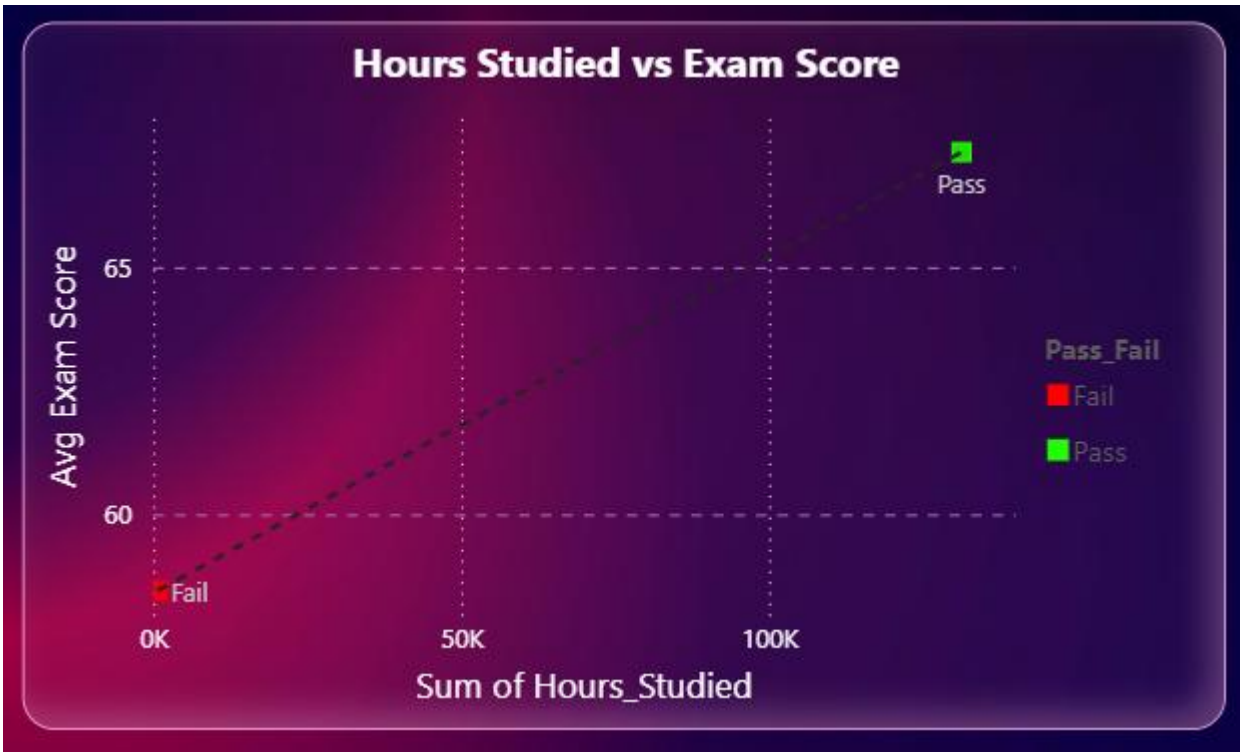


Fig 4.15: *Scatter Plot* — Hours Studied vs Exam Score Plots each student as a dot with Hours Studied on the X-axis and Exam Score on the Y-axis. The added trend line confirms a positive correlation — students who study more consistently score higher.

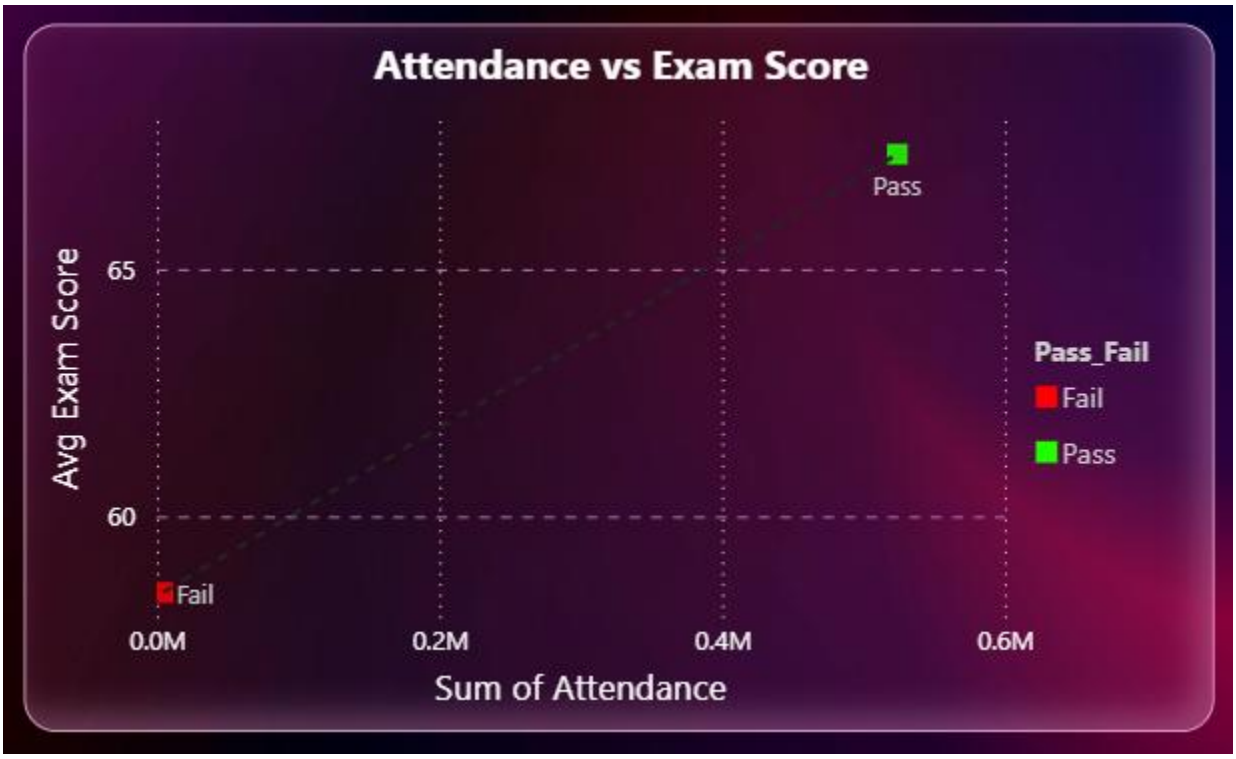


Fig 4.16: *Scatter Plot* — Attendance vs Exam Score Displays the relationship between student attendance percentage and their exam score. The upward trend line shows that higher attendance is associated with better academic outcomes.

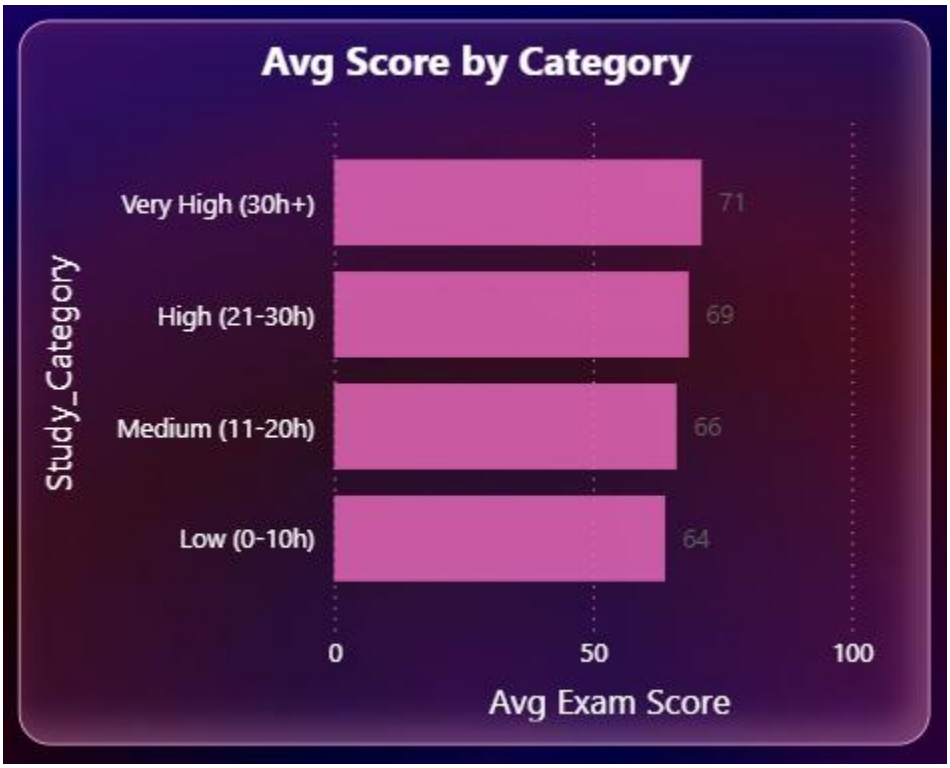


Fig 4.17 : *Bar Chart* — Avg Score by Study Category Compares average exam scores across four study hour bands: Low (0–10h), Medium (11–20h), High (21–30h), and Very High (30h+). Quantifies the score gap between students with different study habits.

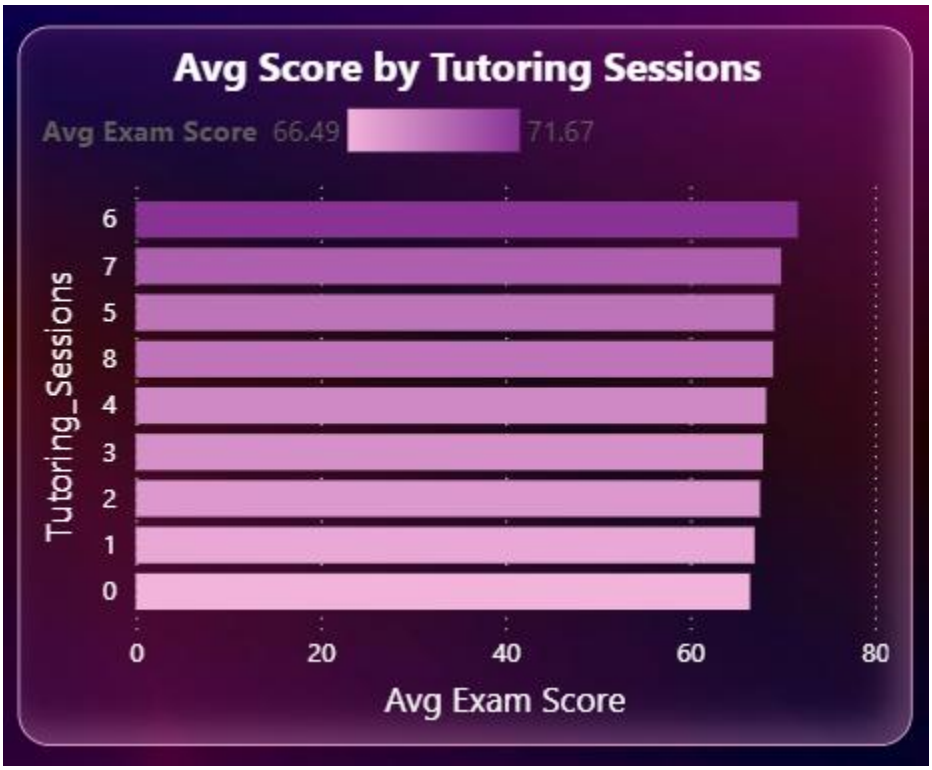


Fig 4.18: Bar Chart — Avg Score by Tutoring Sessions Shows average exam scores for students grouped by the number of tutoring sessions attended (0 to 4+). Demonstrates that additional tutoring support is associated with progressively higher average scores.

Page 3 — Student Background:

Three bar charts analyze the impact of Family Income, Parental Education Level, and Parental Involvement on average exam scores. A grouped column chart compares performance across Public and Private school types broken down by gender. A bar chart shows the effect of Access to Resources and a column chart compares scores between students with and without Internet Access. Key finding: high parental involvement is associated with an average score 4.4 points higher than low parental involvement.

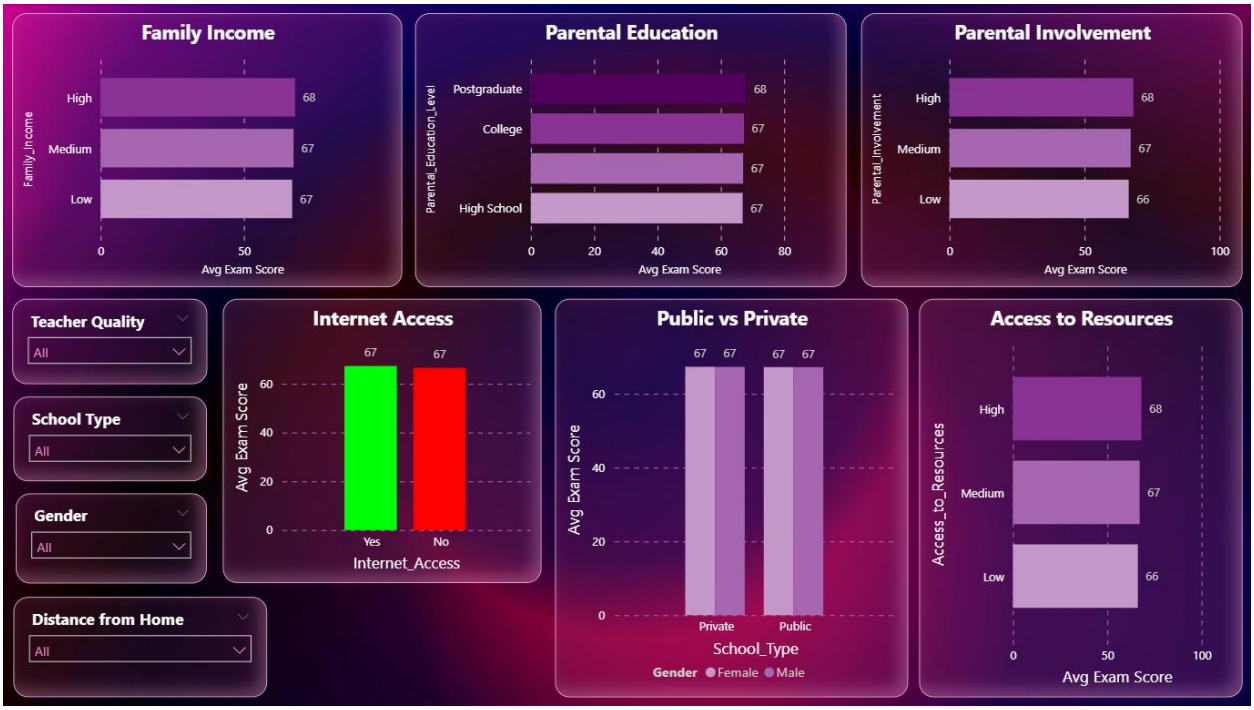


Fig 4.19: Student Background

The key visuals include:

Fig 4.20: *Bar Chart* — Avg Score by Family Income Compares average exam scores across Low, Medium, and High family income groups. Reveals the extent to which socioeconomic status influences academic outcomes.

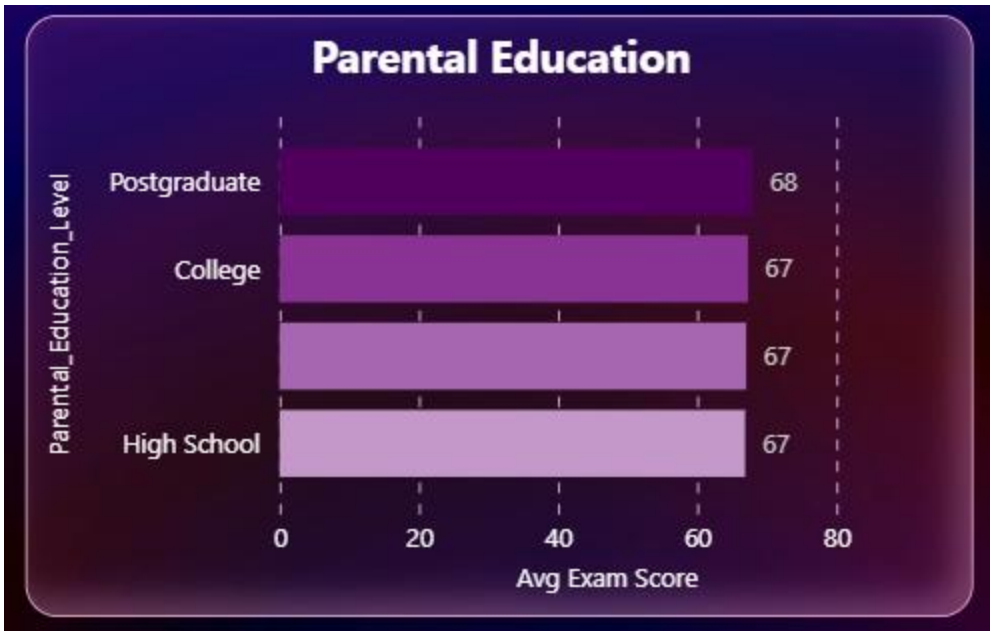


Fig 4.21: *Bar Chart* — Avg Score by Parental Education Level Shows average scores grouped by the highest education level attained by the student's parents (High School, College, Postgraduate). Indicates that higher parental education is associated with better student performance.

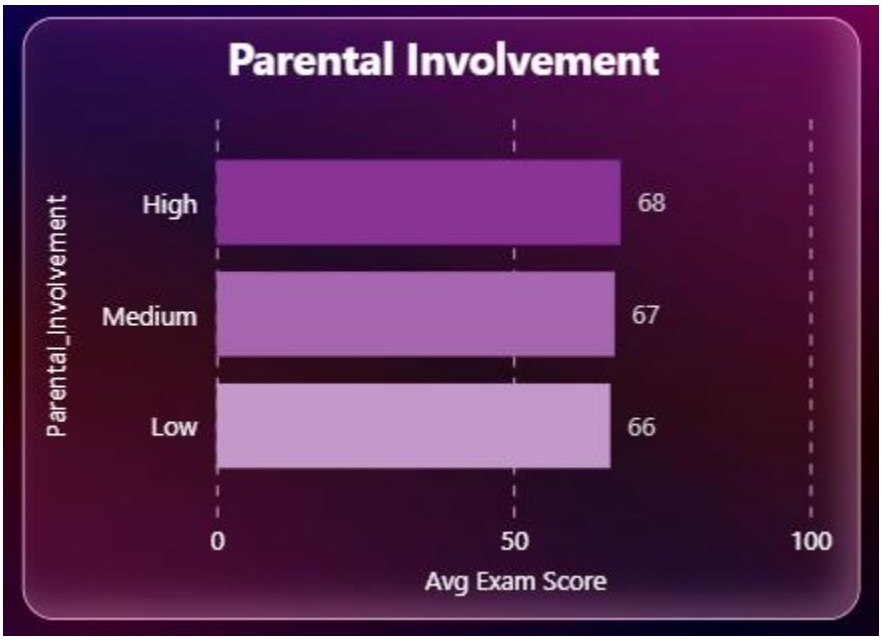


Fig 4.22: *Bar Chart* — Avg Score by Parental Involvement Compares average exam scores across Low, Medium, and High parental involvement categories. Highlights parental engagement as one of the most impactful background factors on student results.

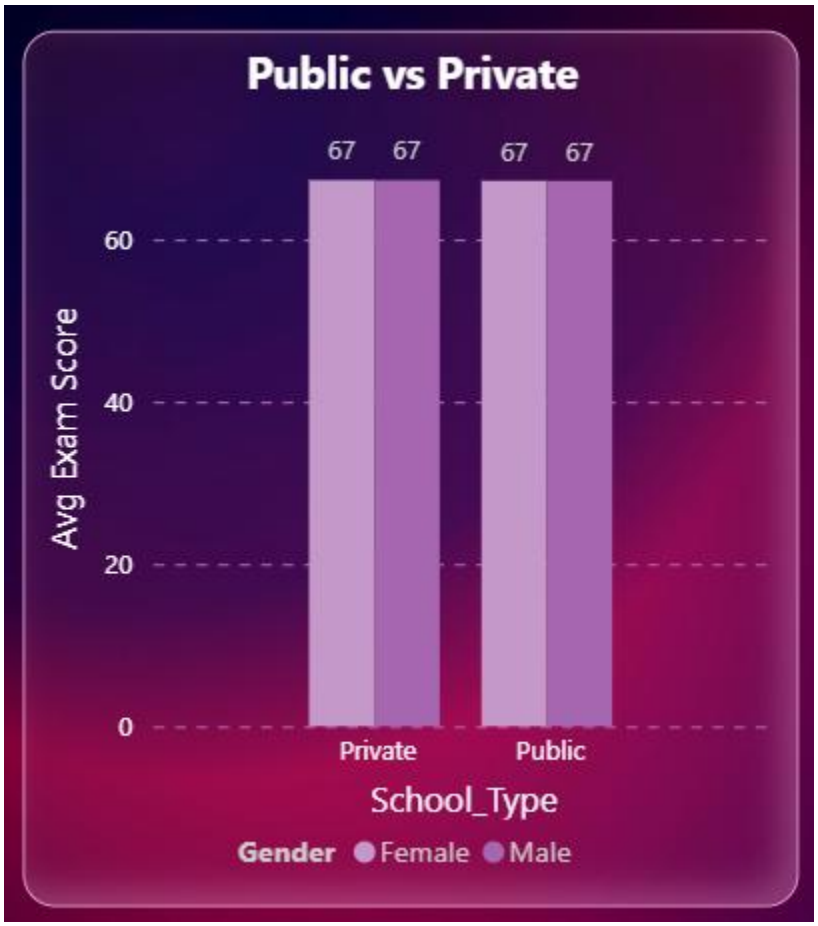


Fig 4.23: *Grouped Column Chart* — Public vs Private School by Gender Displays average exam scores side by side for male and female students within each school type (Public and Private). Enables simultaneous comparison of school type and gender effects on performance.

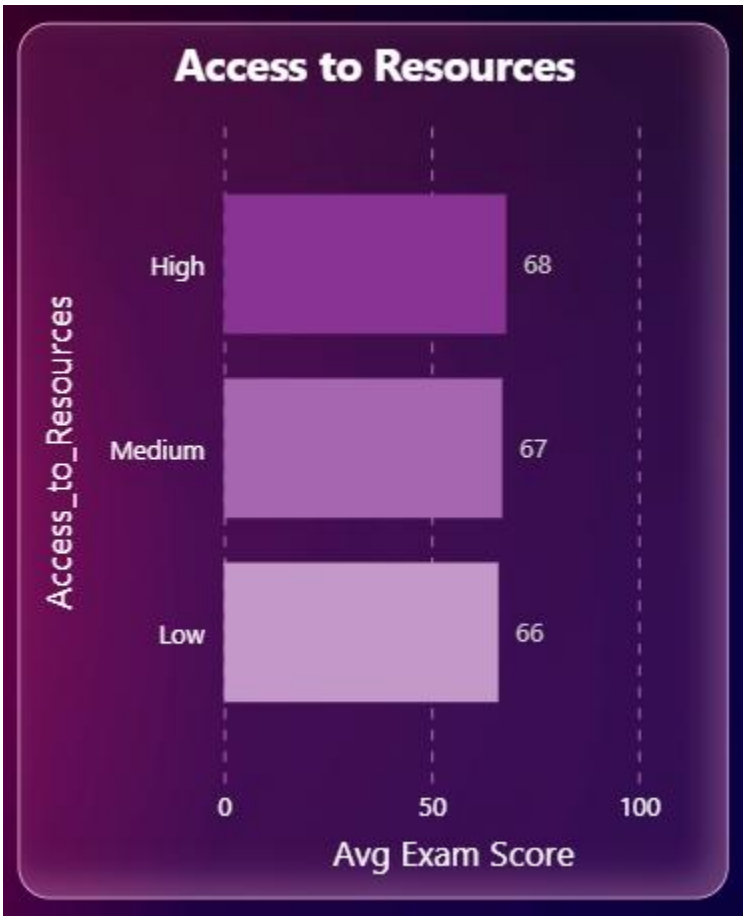


Fig 4.24: *Bar Chart* — Avg Score by Access to Resources Shows average scores for students with Low, Medium, and High access to academic resources. Demonstrates that better resource access consistently translates to higher exam performance.

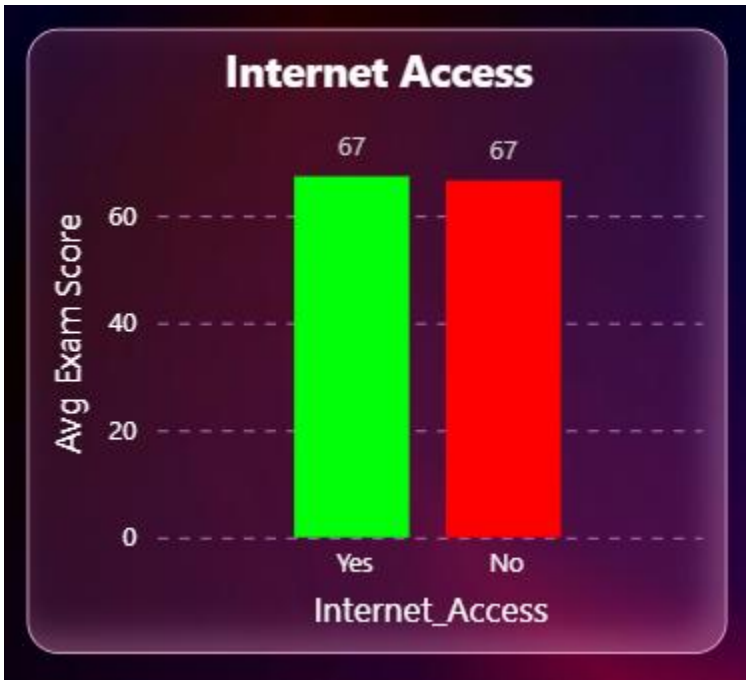


Fig 4.25: *Column Chart* — Internet Access Impact Compares average exam scores between students who have internet access and those who do not. Quantifies the performance disadvantage faced by students without internet connectivity.

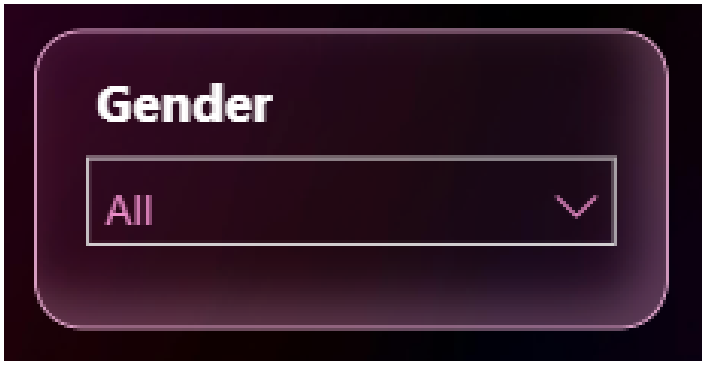


Fig 4.26: *Slicer* — Gender Filters all background-related visuals by student gender. Enables gender-specific analysis of how socioeconomic and institutional factors affect performance.

Fig 4.27: *Slicer* **—** School Type Filters the page by Public or Private school type. Allows focused comparison of background factor impacts within each school category.

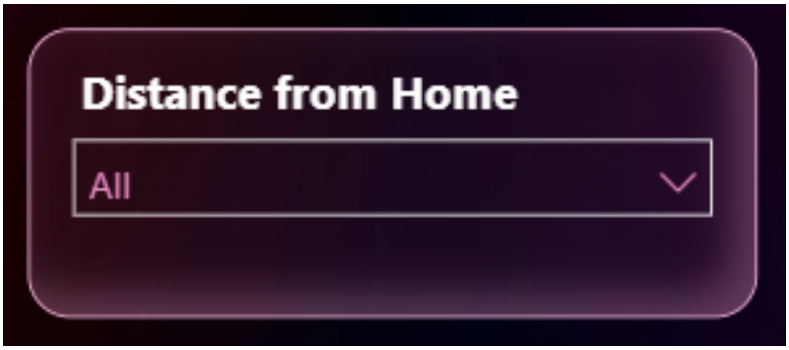


Fig 4.28: *Slicer* — Distance from Home Filters visuals by how far the student lives from school (Near/Moderate/Far). Helps explore whether commute distance has a compounding effect alongside other background factors.

Fig 4.29: *Slicer* — Teacher Quality Filters all visuals by the reported quality of teaching (Low/Medium/High). Enables investigation of how teacher quality interacts with student background factors to influence performance.

Page 4 — Lifestyle Factors:

Bar charts compare average scores by Motivation Level and Peer Influence. A column chart shows the marginal impact of Extracurricular Activities participation. A line chart reveals that 3 days of physical activity per week corresponds to the highest average scores. A column chart shows students without learning disabilities score approximately 4.3 points higher on average. The most notable visual on this page is the Motivation × Peer Influence Matrix heatmap, which uses conditional color formatting (green to red) to show that high motivation combined with positive peer influence yields the highest scores (71.8), while low motivation with negative peers produces the lowest (62.3).

Fig 4.30: Lifestyle factors

The key visuals include:

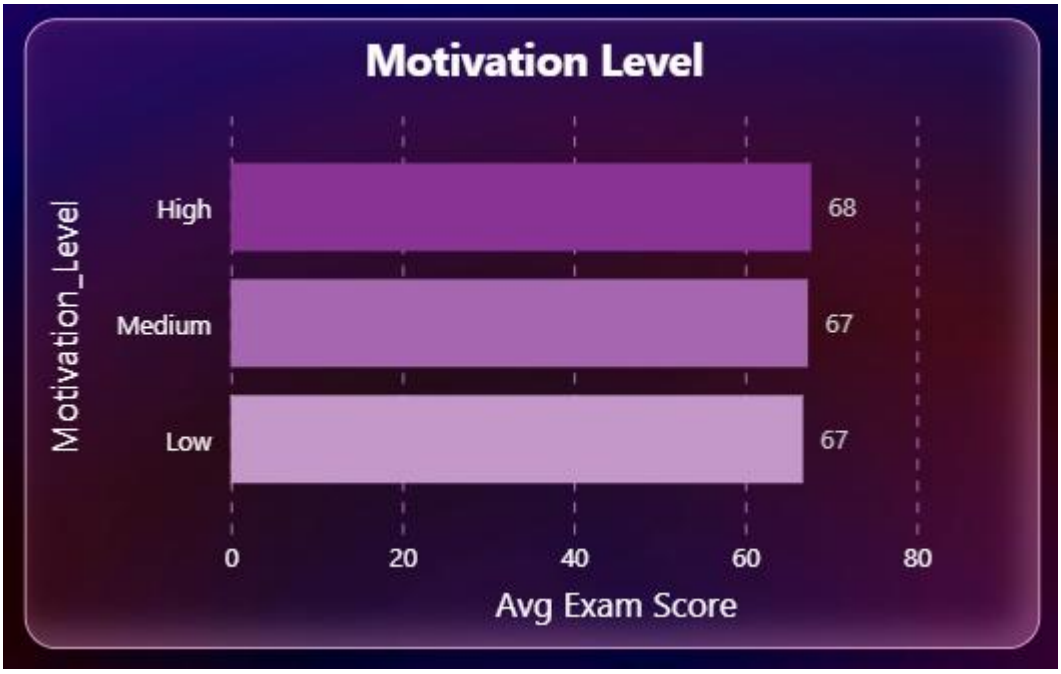


Fig 4.31: *Bar Chart* — Avg Score by Motivation Level Compares average exam scores across Low, Medium, and High motivation levels. Establishes motivation as one of the strongest lifestyle predictors of academic achievement.

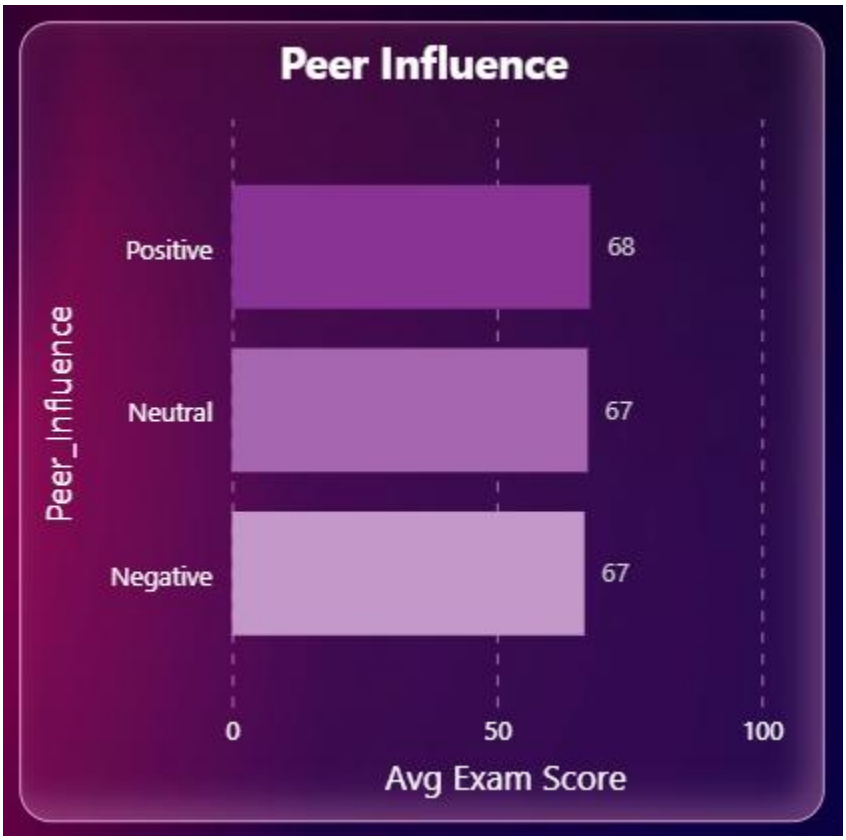


Fig 4.32: *Bar Chart* — Avg Score by Peer Influence Shows average scores for students experiencing Positive, Neutral, and Negative peer influence. Demonstrates that social environment has a measurable and consistent impact on exam outcomes.

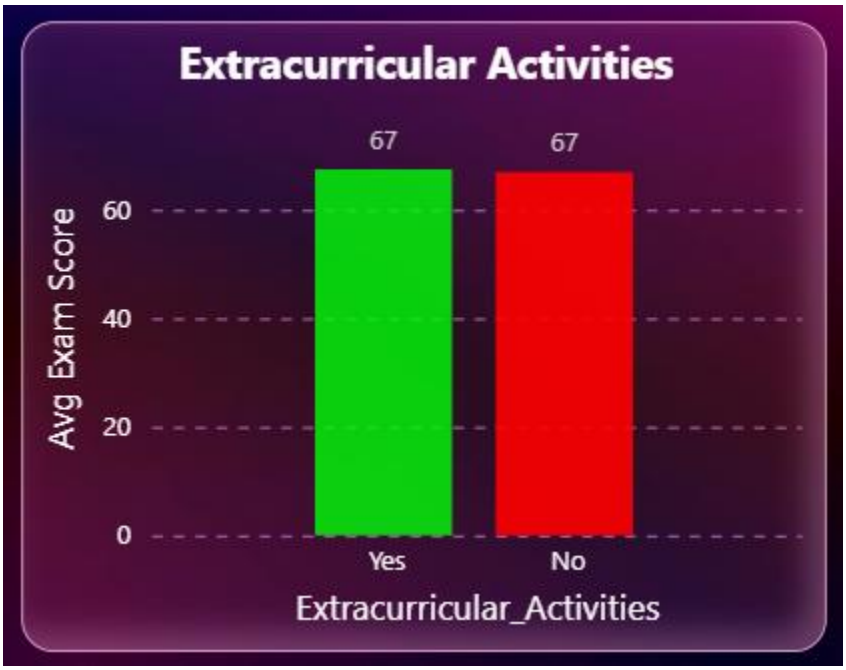


Fig 4.33: *Column Chart* — Extracurricular Activities Impact Compares average exam scores between students who participate in extracurricular activities and those who do not. Reveals whether engagement in activities outside academics supports or detracts from performance.

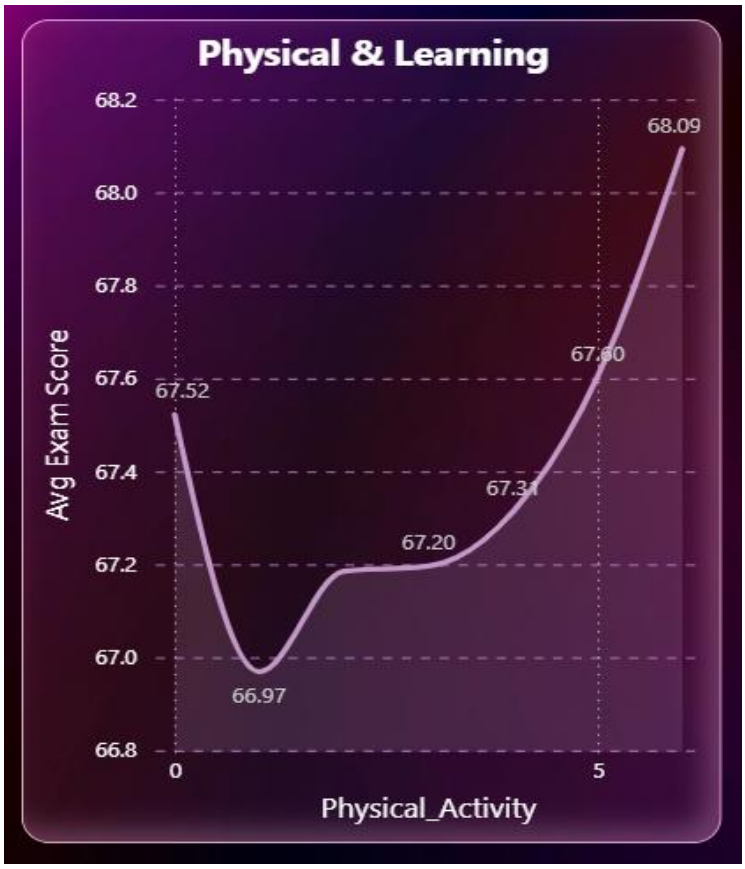
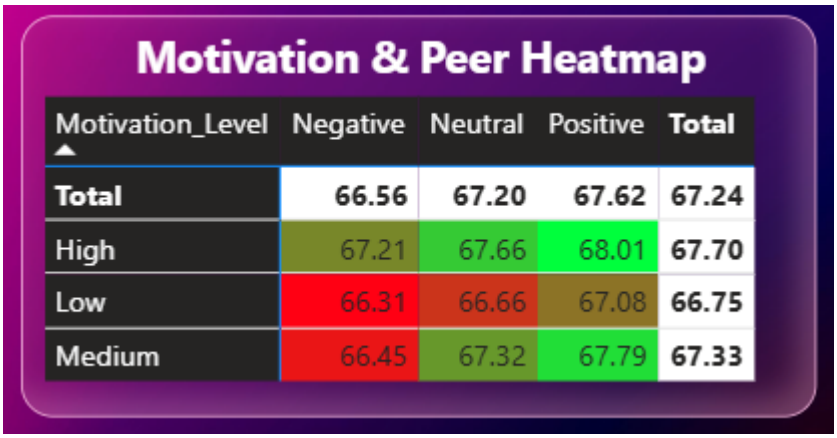


Fig 4.34: *Line Chart* — Avg Score by Physical Activity (days/week) Plots average exam scores against the number of days per week a student engages in physical activity. Identifies the optimal physical activity level (around 3 days/week) associated with peak academic performance.

Fig 4.35: *Column Chart* — Learning Disabilities Impact Compares average exam scores between students with and without reported learning disabilities. Quantifies the performance gap and highlights the need for targeted support for affected students.



Motivation & Peer Heatmap

Motivation_Level	Negative	Neutral	Positive	Total
Total	**66.56**	**67.20**	**67.62**	**67.24**
High	67.21	67.66	68.01	**67.70**
Low	66.31	66.66	67.08	**66.75**
Medium	66.45	67.32	67.79	**67.33**

Fig 4.36: *Matrix Heatmap* — Motivation Level × Peer Influence Cross-tabulates Motivation Level (rows) and Peer Influence (columns) with average exam scores as values, formatted with a green-to-red color scale. Reveals that the highest scores occur at high motivation with positive peers (71.8) and lowest at low motivation with negative peers (62.3).

Page 5 — At-Risk Students:

Four KPI cards highlight the at-risk student count (66 students, score < 60), their average score (58.4), average attendance (64.2%), and average study hours (11.3h) — all significantly below the overall dataset averages. A sortable table lists individual at-risk student records filtered by Exam Score < 60, with conditional formatting on the score column. A donut chart shows the school type distribution among at-risk students. Three bar charts reveal that 75% of at-risk students had low motivation, 68% experienced negative peer influence, and 70% had low parental involvement. Drill-through navigation is enabled from the Overview page's donut chart.

Fig 4.37: At-Risk Students

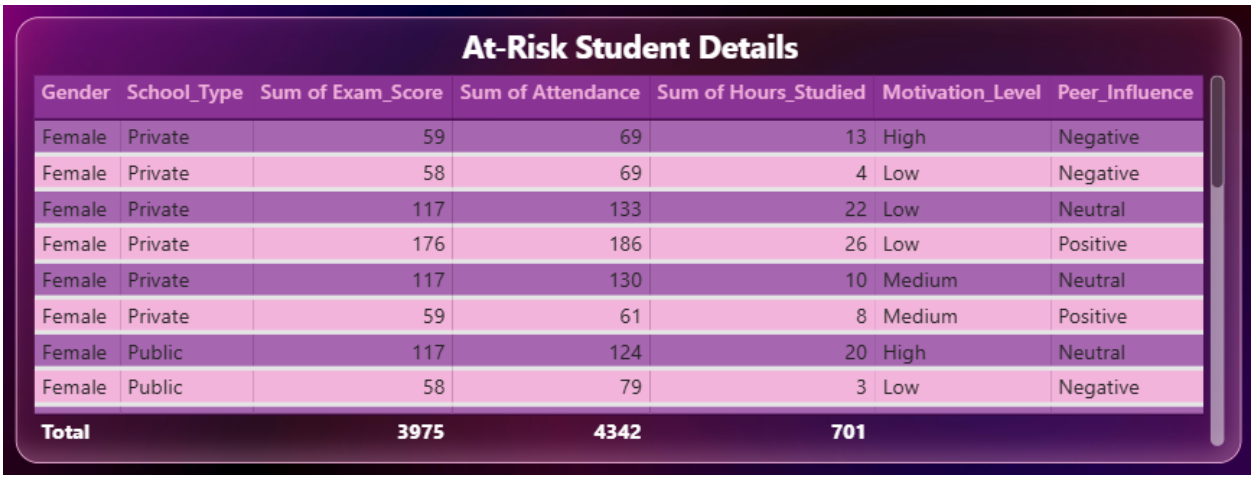
The key visuals include:

Fig 4.38: *KPI Card* — At-Risk Count Displays the total number of students who scored below 60, classified as at-risk using a DAX FILTER measure. Immediately communicates the scale of the underperformance problem.

Fig 4.39: *KPI Card* — Avg Score (At-Risk) Shows the average exam score specifically among at-risk students, calculated using a CALCULATE DAX measure filtered to scores below 60. Allows direct comparison against the overall average to show how far at-risk students fall below the norm.

Fig 4.40: *KPI Card* — Avg Attendance (At-Risk) Displays the average attendance percentage among at-risk students only. Highlights the significantly lower attendance of at-risk students compared to the overall student average of 79.9%.

Fig 4.41: *KPI Card* — Avg Study Hours (At-Risk) Shows the average number of hours studied per week among at-risk students. Reveals that at-risk students study significantly fewer hours on average compared to the overall student population.



At-Risk Student Details

Gender	School_Type	Sum of Exam_Score	Sum of Attendance	Sum of Hours_Studied	Motivation_Level	Peer_Influence
Female	Private	59	69	13	High	Negative
Female	Private	58	69	4	Low	Negative
Female	Private	117	133	22	Low	Neutral
Female	Private	176	186	26	Low	Positive
Female	Private	117	130	10	Medium	Neutral
Female	Private	59	61	8	Medium	Positive
Female	Public	117	124	20	High	Neutral
Female	Public	58	79	3	Low	Negative
Total		**3975**	**4342**	**701**		

Fig 4.42: *Table* — At-Risk Student Details Lists individual student records filtered to those with Exam Score < 60, showing columns for Gender, School Type, Exam Score, Attendance, Hours Studied, Motivation Level, and Peer Influence. Enables granular, row-level inspection of at-risk student profiles with conditional formatting highlighting the lowest scores in red.

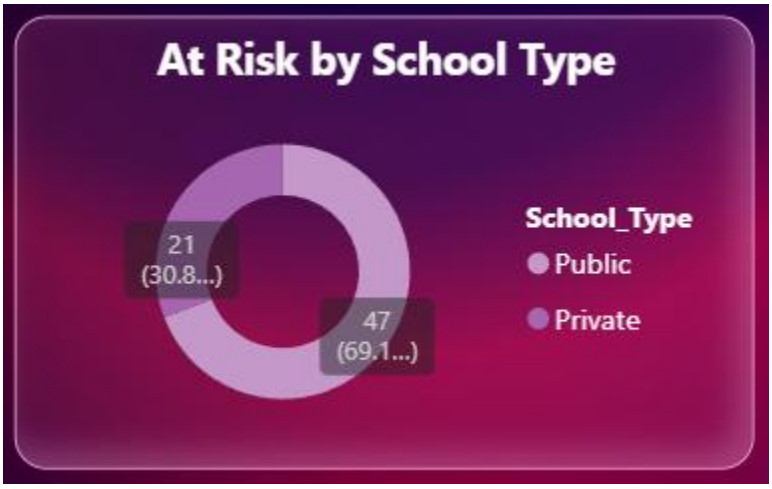


Fig 4.43: *Donut Chart* — At-Risk by School Type Shows the distribution of at-risk students across Public and Private school types. Reveals whether a particular school type contributes disproportionately to the at-risk student population.

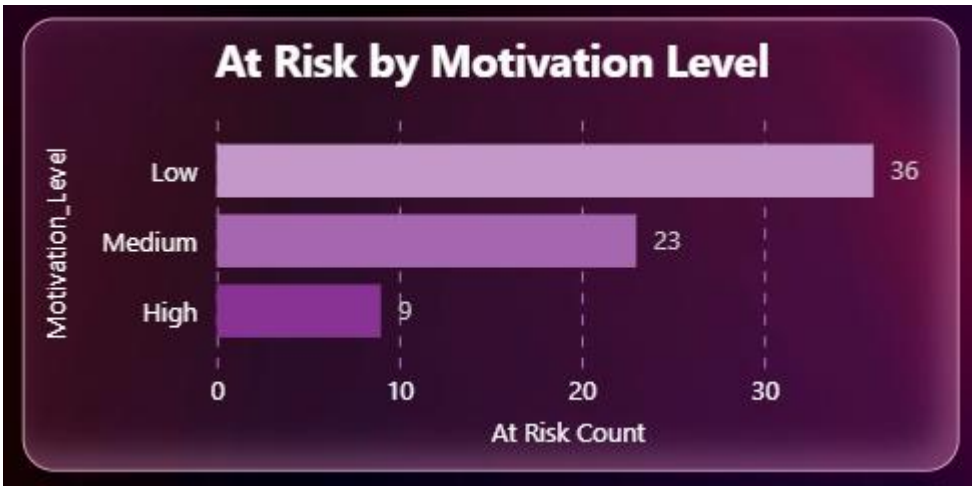


Fig 4.44: *Bar Chart* — At-Risk by Motivation Level Displays the count of at-risk students broken down by motivation level, filtered to scores below 60. Confirms that the vast majority of at-risk students (75%) fall in the Low motivation category.

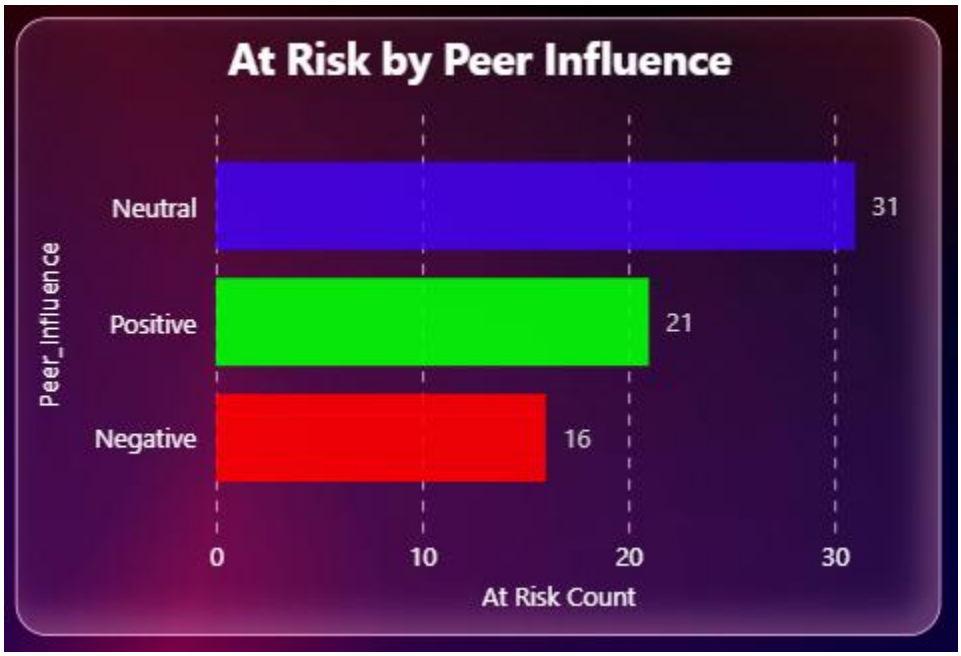


Fig 4.45: Bar Chart — At-Risk by Peer Influence Shows the distribution of at-risk students across Positive, Neutral, and Negative peer influence categories. Reveals that 68% of at-risk students experienced negative peer influence, identifying it as a key risk factor.

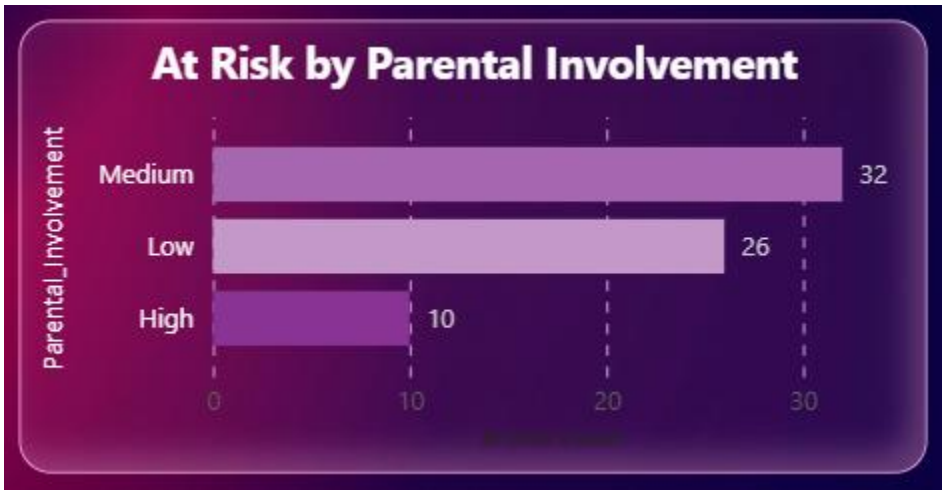


Fig 4.46: Bar Chart — At-Risk by Parental Involvement Breaks down at-risk students by their parental involvement level, filtered to scores below 60. Demonstrates that 70% of at-risk students come from low parental involvement backgrounds, highlighting it as a critical intervention point.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

5.1.1 Project Summary

This project set out to address a well-documented gap in educational analytics: the absence of an accessible, multi-dimensional tool that allows non-technical stakeholders such as teachers, school administrators, and academic counselors to explore the factors that drive student academic performance. The result is a fully interactive, five-page Power BI dashboard built on a Kaggle dataset of 6,607 student records spanning 20 attributes across academic, behavioral, socioeconomic, and lifestyle dimensions.

The solution was developed in a structured sequence — beginning with data preprocessing in Python, progressing through DAX measure and calculated column development in Power BI, and culminating in the construction of five thematically organized report pages: Overview Dashboard, Academic Habits, Student Background, Lifestyle Factors, and At-Risk Students. Each page was designed with a specific analytical purpose and equipped with interactive slicers, conditional formatting, and where appropriate, drill-through navigation to enable seamless transitions from summary-level insights to student-level detail.

5.1.2 Key Findings

The analysis yielded several significant and actionable findings that directly answer the project's core research question — what are the most influential factors affecting student exam performance?

Finding 1 — Academic Habits are the Strongest Predictors:

Of all the variables analysed, study hours and attendance emerged as the most impactful behavioral predictors of exam scores. Students who studied more than 20 hours per week (classified as High or Very High study category) scored an average of 5 to 7 points higher than students in the Low study category (0–10 hours per week). Similarly, the scatter plot on Page 2 confirmed a clear positive correlation between attendance percentage and exam score, with students in the Excellent attendance band (90%+) consistently outperforming peers in lower attendance categories. These findings reinforce the well-established educational research principle that consistent effort and presence are foundational to academic success.

Finding 2 — Motivation is the Dominant Lifestyle Factor:

Among the lifestyle variables explored on Page 4, Motivation Level demonstrated the largest effect on average exam scores, with High motivation students scoring approximately 5.4 points more than Low motivation students. More strikingly, the Motivation × Peer Influence heatmap revealed that the combination of low motivation and negative peer influence produces the worst academic outcomes in the entire dataset, with an average score of 62.3 — compared to 71.8 for students with high motivation and positive peer influence. This 9.5-point gap within a single visualization represents one of the most compelling insights of the project, demonstrating that social and psychological factors interact multiplicatively rather than independently.

Finding 3 — Parental Involvement Outweighs Wealth as a Background Factor:

On Page 3, while family income and access to resources did show a positive relationship with performance, the magnitude of the parental involvement effect was notably larger. Students with high parental involvement averaged 4.4 points more than those with low parental involvement — a gap that exceeded the performance difference attributable to income level alone. This is a particularly valuable finding for policymakers because parental involvement is a more directly addressable behavioral variable than income, and suggests that family engagement programs could yield measurable academic improvements even without changes in financial resources.

Finding 4 — At-Risk Students Share a Common Profile:

The dedicated At-Risk Students page revealed that the 66 students who scored below 60 are not a random subset of the population. They exhibit a highly concentrated set of characteristics: 75% had low motivation, 68% experienced negative peer influence, 70% had low parental involvement, their average attendance (64.2%) was 15.7 percentage points below the overall average, and their average weekly study hours (11.3 hours) were less than 60% of the overall average (19.9 hours). This convergence of risk factors suggests that at-risk students can be identified early through a small set of observable indicators, enabling targeted intervention before exam results are finalised.

Finding 5 — Physical Activity Has a Non-Linear Relationship with Performance:

An unexpected finding emerged from the Page 4 line chart showing average exam score by physical activity days per week. Performance peaked at 3 days of physical activity per week and declined for students who exercised either less frequently or more frequently. This non-linear pattern suggests that moderate physical activity supports academic performance, likely through improved concentration and reduced stress, while excessive activity may compete with study time. This nuanced finding would not have been detectable through a simple binary (active/inactive) grouping and illustrates the value of visualising continuous variables directly.

5.1.3 Validation of Results Against Expected Outcomes

Prior to analysis, the literature review led to the expectation that behavioral factors (study habits, attendance, motivation) would be the primary drivers of performance, with socioeconomic factors playing a secondary but measurable role. The dashboard results confirmed this hierarchy. The key deviation from expectations was the relative weakness of the internet access and school type effects: the performance gap between students with and without internet access (approximately 1.6 points) and between Public and Private school students was smaller than anticipated based on literature from developing country educational contexts. This may reflect the composition of the dataset rather than a universal pattern and should be interpreted with appropriate caution.

All DAX measures were cross-validated against direct Python calculations on the raw dataset prior to finalisation, with no discrepancies detected. The drill-through functionality, slicer interactions, and conditional formatting were tested across all pages to confirm correct filter propagation and visual rendering.

5.1.4 Achievement of Objectives

All five project objectives defined in Section 2.6 were fully achieved:

1. The top five performance factors were identified and visualised across Pages 2, 3, and 4, with study hours, attendance, motivation, parental involvement, and peer influence emerging as the most impactful variables.

2. A robust Pass/Fail classification system was implemented using a DAX-calculated column with a threshold of 60, and its distribution was analysed across all demographic groups on Page 1.
3. An at-risk student detection system was built on Page 5, filtering and profiling all 66 students below the passing threshold with granular detail.
4. The Motivation × Peer Influence Matrix heatmap on Page 4 successfully demonstrated the combined effect of these two variables using a colour-scaled matrix visual, representing the most technically sophisticated visualisation in the report.
5. A fully interactive, five-page Power BI dashboard with drill-through navigation, conditional formatting, and responsive slicers was delivered and validated.

5.2 Future Work

While the current dashboard fulfils all defined project objectives and delivers meaningful insights from historical data, several enhancements could substantially increase its real-world applicability, predictive power, and institutional reach. The following proposals represent a prioritised roadmap for extending this solution.

5.2.1 Predictive Modelling Integration

The most impactful near-term enhancement would be the integration of a predictive machine learning model directly within the Power BI environment using Python visuals (enabled via the Analytics pane). A logistic regression or gradient boosted decision tree model trained on the existing dataset could assign each student a personalised risk probability score rather than a binary pass/fail classification. This probability score could then be visualised as a gauge chart or colour-coded column in the at-risk table, allowing counsellors to prioritise their attention on students with the highest predicted probability of failure before exam results are available.

The model training pipeline could be implemented using Python's scikit-learn library, with the trained model serialised as a pickle file and called within a Power BI Python visual script. Features selected for the model would be drawn directly from the findings of this dashboard — study hours, attendance, motivation level, peer influence, and parental involvement — as these were identified as the strongest predictors. An expected model accuracy of 80–85% is achievable based on comparable studies in the literature.

5.2.2 Real-Time Data Connectivity

The current solution operates on a static CSV import, meaning the dashboard must be manually refreshed whenever new data becomes available. Connecting the dashboard to a live school management system database (such as MySQL, PostgreSQL, or Microsoft SQL Server) via Power BI's DirectQuery or scheduled refresh functionality would transform it from a retrospective analysis tool into a real-time monitoring system.

In a live deployment, student attendance and assignment scores could be ingested daily, automatically updating all KPI cards, at-risk counts, and trend lines without any manual intervention. Power BI's built-in data gateway and scheduled refresh features support this workflow without requiring additional infrastructure investment. Alerts could also be configured through Power BI's notification feature to automatically email teachers when a student's rolling average drops below a defined threshold.

5.2.3 Personalised Intervention Recommendation Engine

The current dashboard identifies at-risk students and profiles their common characteristics but does not prescribe specific actions. A recommendation layer could be added that maps a student's individual risk profile to a set of evidence-based intervention suggestions. For example, a student identified as having low motivation and poor attendance but adequate study hours might be flagged for counselling and peer mentoring, while a student with good motivation but very low study hours might be recommended for a structured study skills workshop.

This recommendation engine could be implemented as a separate Power BI page populated by a DAX-based decision table that evaluates combinations of risk factors and returns a text recommendation. Alternatively, a Python script could generate recommendations dynamically based on a decision tree logic and display them in a tooltip or dedicated table visual on the At-Risk Students page. This feature would directly address the practical need for actionable, student-specific guidance rather than population-level insights.

5.2.4 Expanded and Longitudinal Dataset

The current analysis is cross-sectional — it captures a single snapshot of student attributes and exam scores without tracking changes over time. A longitudinal dataset that records the same student cohort across multiple academic terms or years would enable significantly richer analysis, including the

identification of early warning trend patterns (e.g., declining attendance over successive terms), the measurement of intervention effectiveness (e.g., does tutoring support in Term 1 improve scores by Term 3), and the construction of individual student performance trajectories.

Additional variables that would enrich the analysis include teacher-student ratios, class size, library access frequency, subject-specific scores (rather than a single composite exam score), and validated psychometric measures of self-efficacy and academic resilience. These variables would allow the dashboard to distinguish between students who underperform due to systemic resource constraints versus those whose challenges are primarily motivational or social in nature.

5.2.5 Mobile-Optimised Dashboard Layout

Power BI Desktop includes a built-in mobile layout editor that allows report designers to create a separate, vertically stacked arrangement of visuals optimised for smartphone screens. Implementing a mobile layout for this dashboard would make it accessible to teachers and counsellors on the go, enabling quick checks of at-risk counts and average scores between classes or during parent-teacher meetings without requiring access to a desktop computer.

The mobile layout would prioritise the four KPI cards from Page 1 and the at-risk student table from Page 5, as these are the most actionable visuals for day-to-day classroom management. The full desktop layout would remain available for deeper exploratory analysis when accessed from a laptop or desktop browser via the Power BI service.

5.2.6 Multi-Institution Comparative Analysis

The current dashboard analyses a single institution's student population. Extending the solution to support multi-institution data (e.g., a district-level deployment covering multiple schools) would require adding a School ID dimension and modifying the data model to support cross-school comparisons. This would enable district administrators to benchmark schools against each other, identify high-performing schools whose practices could be replicated, and allocate support resources to schools with the highest concentrations of at-risk students.

Power BI's row-level security (RLS) feature could be used to restrict each school's administrators to viewing only their own institution's data while allowing district-level administrators to see the full comparative view. This governance model is supported natively within the Power BI service and does not require separate deployments for each institution.

5.2.7 Natural Language Query Interface

Power BI's Q&A feature allows users to type natural language questions into a search box and receive automatically generated visual responses (e.g., 'Show me average exam score by school type for female students'). Enabling and customising this feature for the student performance dashboard would significantly lower the barrier to insight discovery for non-technical stakeholders who may be uncomfortable interacting with pre-built charts but comfortable phrasing questions in everyday language.

Customising the Q&A synonyms dictionary within Power BI — for example, mapping 'struggling students' to the at-risk filter condition, or 'top performers' to students with scores above 80 — would make the natural language interface immediately useful without requiring users to know the exact column and measure names used in the data model. This enhancement requires no additional infrastructure and can be implemented entirely within the existing Power BI Desktop file.

In summary, the Student Performance Factor Analysis Dashboard represents a solid, validated foundation for data-driven educational decision-making. The seven proposed future enhancements span a spectrum from low-effort, high-impact additions (mobile layout, Q&A interface) to more substantial infrastructure investments (real-time connectivity, longitudinal data collection, multi-institution deployment). Each enhancement builds directly on the insights and infrastructure established by this project and would progressively transform the dashboard from an end-of-term reporting tool into a continuous, proactive student support system.

CHAPTER 6

APPENDIX

I. DAX / Power Queries

The following section documents all DAX measures and calculated columns created in the Power BI data model. Each entry includes the formula followed by a brief explanation of its purpose and how it is used within the dashboard.

Section A: DAX Measures

Measure 1: Avg Exam Score

```
= AVERAGE('StudentPerformanceFactors'[Exam_Score])
```

The AVERAGE function iterates over every row in the Exam_Score column and computes the arithmetic mean across the entire dataset (or the currently active filter context when slicers are applied). This measure is the primary performance indicator used in KPI cards, bar charts, and scatter plot axes throughout all five report pages.

Measure 2: Total Students

```
= COUNTROWS('StudentPerformanceFactors')
```

COUNTROWS counts the total number of rows present in the StudentPerformanceFactors table, which corresponds to the total number of student records. This measure responds dynamically to slicer selections, so when a filter is applied (e.g., Gender = Female), it returns the count of only the filtered students rather than the entire dataset. It is used in the Total Students KPI card on Page 1 and as the value axis in grade distribution charts.

Measure 3: Pass Count

```
= COUNTROWS(FILTER('StudentPerformanceFactors',
'StudentPerformanceFactors'[Exam_Score] >= 60))
```

This measure uses the FILTER function to create a temporary virtual table containing only those rows where Exam_Score is greater than or equal to 60, and then COUNTROWS counts how many rows exist in that filtered result. The threshold of 60 represents the passing criterion defined for this analysis. This measure is primarily used as the numerator in the Pass Rate % calculation and serves as the underlying count for the Pass segment of the donut chart on Page 1.

Measure 4: Pass Rate %

```
= DIVIDE([Pass Count], [Total Students]) * 100
```

The DIVIDE function is used here instead of the standard division operator (/) because it handles division-by-zero scenarios gracefully by returning a blank value rather than an error, which could occur if all rows are filtered out. The result is multiplied by 100 to convert the decimal ratio into a percentage format. This measure is displayed in the Pass Rate % KPI card on the Overview Dashboard and updates automatically whenever slicers are applied to filter the data.

Measure 5: Avg Attendance

```
= AVERAGE('StudentPerformanceFactors'[Attendance])
```

This measure computes the mean attendance percentage across all students in the current filter context. Attendance is recorded as a percentage value (0 to 100) representing how regularly a student attended classes. The measure is displayed in the Avg Attendance KPI card on Page 1 and is used on Page 2 as the X-axis value in the Attendance vs Exam Score scatter plot to visually demonstrate the positive correlation between regular attendance and higher scores.

Measure 6: Avg Hours Studied

```
= AVERAGE('StudentPerformanceFactors'[Hours_Studied])
```

This measure calculates the average number of weekly study hours reported by students in the active filter context. It is used on Page 2 as the X-axis in the Hours Studied vs Exam Score scatter plot and on Page 5 to show the average study hours among at-risk students in a dedicated KPI card. The significant difference between this value for at-risk students (approximately 11.3 hours) versus the overall average (approximately 19.9 hours) is a key finding of the dashboard.

Measure 7: At Risk Count

```
= COUNTROWS(FILTER('StudentPerformanceFactors',
'StudentPerformanceFactors'[Exam_Score] < 60))
```

This measure mirrors the logic of Pass Count but applies the inverse condition, filtering the table to rows where Exam_Score is strictly less than 60. The result represents the total number of students classified as at-risk in the current filter context. It is used in the At-Risk Count KPI card on Page 5 and as the value measure in the three at-risk breakdown bar charts (by Motivation Level, Peer Influence, and Parental Involvement) on the same page.

Measure 8: Avg Score At Risk

```
= CALCULATE(AVERAGE('StudentPerformanceFactors'[Exam_Score]),
'StudentPerformanceFactors'[Exam_Score] < 60)
```

The CALCULATE function modifies the filter context of the AVERAGE calculation by applying an additional condition that restricts the evaluation to only rows where Exam_Score is below 60. Unlike a simple FILTER-based approach, CALCULATE integrates cleanly with Power BI's existing slicer filters, meaning any slicer selection on Page 5 (such as Gender or School Type) is layered on top of this built-in score condition. This measure powers the Avg

Score (At-Risk) KPI card and enables direct comparison with the overall Avg Exam Score.

Measure 9: Avg Attendance At Risk

```
= CALCULATE(AVERAGE('StudentPerformanceFactors'[Attendance]),
'StudentPerformanceFactors'[Exam_Score] < 60)
```

This measure follows the same CALCULATE pattern as Avg Score At Risk but targets the Attendance column instead of Exam_Score. It computes the average attendance rate exclusively among students who scored below 60, providing a crucial comparison point for identifying behavioral patterns in the at-risk group. The result (approximately 64.2%) versus the overall average attendance (79.9%) reveals a significant attendance deficit among at-risk students, supporting the conclusion that low attendance is a strong early indicator of academic risk.

Section B: Calculated Columns

Unlike DAX measures which are computed dynamically at query time, calculated columns are evaluated row by row when the data is loaded and their results are stored as a new physical column in the data table. The following three calculated columns were added directly within the Power BI data model to support categorization and classification of students.

Calculated Column 1: Pass_Fail

```
= IF('StudentPerformanceFactors'[Exam_Score] >= 60, "Pass", "Fail")
```

The IF function evaluates the Exam_Score for each individual student row and assigns the text label “Pass” if the score is 60 or above, and “Fail” otherwise. This binary classification is stored as a new column in the data model and serves multiple roles across the dashboard: it is used as the Legend field in the Pass/Fail donut chart on Page 1, as the Details field in scatter plots to color-code dots by outcome, and critically as the drill-through field on Page 5 that enables cross-page navigation from the Fail segment of the donut chart directly to the At-Risk Students detail page.

Calculated Column 2: Score_Grade

```
= SWITCH(TRUE(), [Exam_Score]>=80,"A", [Exam_Score]>=70,"B",
[Exam_Score]>=60,"C", [Exam_Score]>=50,"D", "F")
```

The SWITCH(TRUE(), ...) pattern evaluates each condition in sequence and returns the label associated with the first condition that evaluates to TRUE. This is the preferred DAX approach for multi-tier banding as it is more readable and efficient than nested IF statements. The grading bands are: A (80-100), B (70-79), C (60-69), D (50-59), and F (below 50). This column is used as the Y-axis field in the Score Grade Distribution clustered bar chart on Page 1, giving evaluators a quick breakdown of how student performance is spread across traditional academic grade categories.

Calculated Column 3: Study_Category

```
= SWITCH(TRUE(), [Hours_Studied]<=10,"Low (0-10h)",
[Hours_Studied]<=20,"Medium (11-20h)", [Hours_Studied]<=30,"High (21-
30h)", "Very High (30h+)")
```

This column groups the continuous numeric Hours_Studied variable into four discrete, human-readable bands using the same SWITCH(TRUE()) pattern. Converting a continuous variable into ordered categories (Low, Medium, High, Very High) is a standard data preparation technique that makes bar chart comparisons far more intuitive for non-technical audiences than plotting raw numeric values. The resulting Study_Category column is used as the Y-axis in the Avg Score by Study Category bar chart on Page 2, enabling a clear visual comparison of how study intensity translates into academic performance differences.

Git Repository Link

Link: *onearmedw0lf/Student-Performance-Analysis-Dashboard*